

From Brain Scores to Brain-Guided Distillation: Testing Brain Alignment as a Training Signal

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Abstract

Brain alignment measures whether a model representation can predict recorded human brain responses to the same language stimuli on held-out examples. Successful prediction establishes linearly accessible brain-relevant information under a specified readout and control set, but it does not establish a shared biological mechanism. This thesis asks whether recorded or predicted neural responses can also provide useful auxiliary supervision when training a smaller language model by knowledge distillation. The experiments optimize neural-target losses rather than the encoding score itself, then evaluate the retained students with held-out, nuisance-controlled brain encoding. Trained language-model features contain measurable signal beyond the specified nuisances, but the tested objectives based directly on recorded fMRI do not establish a participant-level benefit at comparable language-model quality. With a fresh ridge readout fitted after training, TRIBE-supervised students achieved held-out synthetic-target $R^2 + 0.078298$ higher than seed-matched KD students [E016][E026]. Their retained parameters and representations also changed while language quality remained within the predeclared envelope [E025][E026]. The saved text-feature target was not a valid control for attributing that difference to neural target content. TRIBE’s KD baseline left 4.582 times as much remaining headroom as the text-feature KD baseline. The targets differed in covariance geometry and nuisance predictability, and the trained arms differed in parameter displacement and centered-Frobenius representation change [E026]. Across nine participants, the mean direct contrast in held-out unique R^2 , TRIBE minus KD, was -0.000014 , with a 95% CI of $[-0.000739, +0.000711]$ [E025]. The predeclared primary mean contrast, TRIBE minus textfeat, was $+0.000136$ unique R^2 , with a 95% CI of $[-0.000505, +0.000777]$. Its one-sided 95% upper confidence bound was $+0.000653$, below the internal $+0.002$ continuation threshold [E025]. Because the comparator was not valid for target-content attribution, this relative contrast cannot isolate an effect of neural target content [E026]. These results separate target uptake, retained-model movement, comparator identification, and participant-level biological transfer, and they do not demonstrate a reliable brain-guided training benefit in the tested distillation regimes.

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1 Introduction

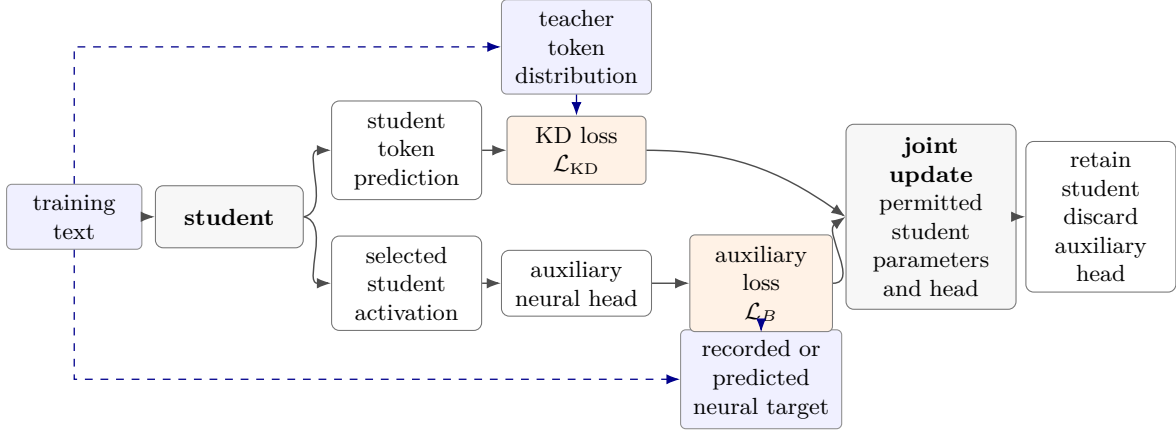
Imagine that a person listens to a story in an fMRI scanner while a language model processes the same words. For each part of the story, the scanner provides a vector of measured responses across locations in the brain, while a chosen layer of the model provides a vector of artificial-neuron activations. An *encoding model* learns a linear map from the model activations to the brain responses using some of the story, and is then evaluated on parts of the story that were not used to fit the map. If the map predicts held-out responses, distinctions that matter for the measured brain signal are linearly accessible in the model representation. This predictive correspondence is what this thesis calls *brain alignment*.

The construction is easier to understand through one concrete example. Suppose the person hears “the child opened the present” and later hears “the mechanic repaired the engine.” The model produces different activation patterns because the words, syntax, and meanings differ, while the scanner records different patterns across language-responsive locations. The encoding map is fitted across many such examples and is required to predict a brain-response pattern for a sentence or story segment it did not see during fitting. It succeeds only when the geometry of the model features contains distinctions that a simple linear readout can reuse for the recorded responses. The map is therefore a measurement instrument between two representational spaces, not a component that either the model or the brain is assumed to contain.

The phrase “predict the brain” also needs calibration. An R^2 score does not reconstruct thoughts, identify what a participant consciously understood, or imply that every measured voxel is language selective. It summarizes how well one predefined response target can be predicted relative to a baseline on held-out stimuli. Changing the participant pool, averaging responses before scoring, selecting a different ROI, or using a different split can change both the numerical score and the scientific population it describes. For that reason, the experimental design is part of the claim rather than a technical detail hidden behind the score.

Brain alignment is scientifically interesting because it offers a common measurement for two otherwise very different systems. Researchers can ask which model layers best predict language-selective brain responses, whether training improves the correspondence, and whether scaling or compressing a model preserves it [1–4]. The score is nevertheless narrower than its name can suggest. Successful held-out prediction does not show that a model thinks like a person, implements the same computation, or contains a generally useful cognitive representation. It shows that a specified readout extracts predictive information under a specified dataset, nuisance model, split, and inference unit. Low-level stimulus features, temporal leakage, response

Intervention



Evaluation with retained student weights

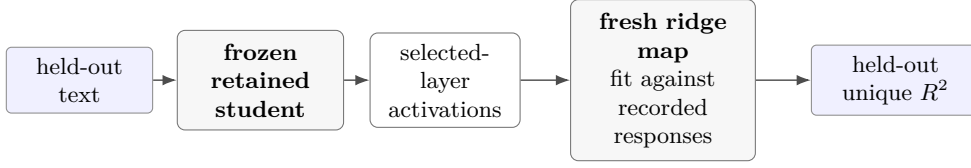


Figure 1: Training and evaluation use different quantities. For the same training text, the KD loss compares the student’s token prediction with the teacher distribution. The auxiliary loss compares the neural head’s prediction with the recorded or predicted neural target. Both losses update permitted student parameters; only the auxiliary loss updates the neural head. That head is discarded after training. We then freeze the retained student, fit a fresh nuisance-controlled ridge encoding map, and report held-out unique R^2 .

averaging, and language-model quality can all change the score without establishing the stronger interpretation [5–7].

This thesis tests whether neural response targets can improve a smaller language model. In knowledge distillation, a smaller *student* model learns from the output distribution of a larger *teacher* model. Distillation can retain much of the teacher’s predictive quality while changing the student’s internal representation. For example, when the teacher assigns nonzero probability to several plausible next words, the student learns from that full probability pattern rather than only from the observed next token. This transfers the teacher’s output behavior, but it does not require the student’s middle layers to organize information in the same way. Brain-guided distillation adds a second training signal that rewards agreement with a recorded or predicted neural target at a selected student layer. The training objective is a neural-target loss, such as mean-squared error, not the held-out encoding score used to evaluate the retained student. The comparison of interest is not simply whether the resulting student has a higher brain score than an arbitrary baseline. It is whether neural-target supervision improves held-out alignment relative to an appropriate nonbrain control while teacher, student, text data, training budget, and achieved language-model quality are held as comparable as the design permits. The central question is whether a recorded or predicted neural target improves a smaller student and whether that change transfers to held-out recorded brain responses.

Figure 2 organizes the experiments into five requirements for progressively stronger claims: controlled measurement, preservation and quality, an identified intervention, biological transfer, and external utility. Gate 3 expands the intervention check into target uptake, retained-student change, language-quality preservation, and comparator validity. The ordering governs interpretation, but a later diagnostic can still locate a failure boundary after an earlier condition fails. The checklist organizes the experiments; staged validation itself is not a claimed contribution.

A positive raw score without nuisance and untrained-network controls can reflect stimulus structure rather than learned language processing. A lower score after compression can reflect a worse language model rather than loss caused by distillation itself. A student representation can become more predictive of its auxiliary target without gaining structure that transfers to recorded brain responses. Two arms can also pass separate manipulation checks while differing in baseline target difficulty, loss scale, or the amount of model change. Agreement across training seeds does not establish consistency across participants because seeds and participants index different sources of variation. Finally, improvement on the same proxy used for training can fail to transfer to recorded fMRI or to a downstream language task. The analyses therefore report these conditions separately.

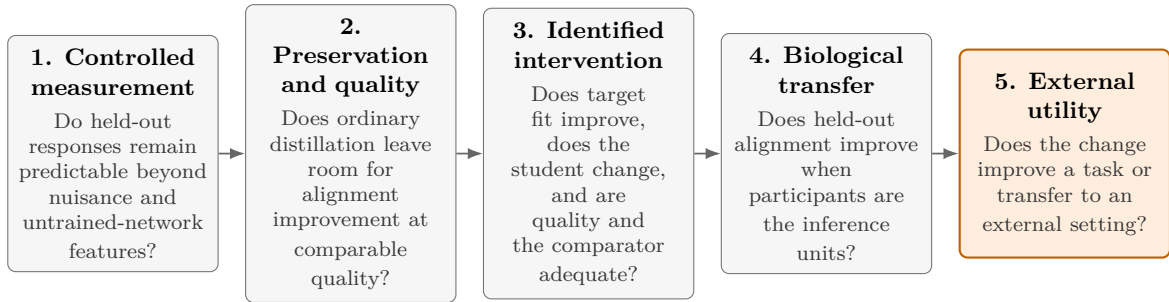


Figure 2: The five stages used to organize the experimental program. The intervention stage includes target uptake, retained-student movement, language-quality preservation, and comparator validity. Biological transfer then requires evaluation on recorded responses at the participant inference unit before any external utility claim is made.

The motivating hypothesis is deliberately weak. A brain objective may act as a regularizer that selects among text-compatible student representations rather than as a second teacher that supplies a new language task. This possibility is plausible because next-token objectives constrain output distributions more directly than middle-layer geometry, leaving multiple internal solutions with similar perplexity. It is not guaranteed by information theory, and the brain dataset’s small and noisy supervision does not by itself establish a useful information budget. The hypothesis applies most plausibly to constrained, compressed, or under-trained regimes in which an auxiliary objective could select among text-compatible solutions. The experiments test whether such selection occurs and whether it transfers beyond the optimized target.

The work is positioned between four related literatures. Encoding-model studies measure correspondence between artificial and biological representations [1–3]. Neural-guided optimization studies use recorded or predicted responses to change learned representations [8–10]. Compression studies ask how pruning, quantization, or distillation changes alignment after training [4]. Validity studies show why nuisance subtraction, reliable splits, matched quality, and the correct biological inference unit are necessary before a score is interpreted [5–7]. The contribution is not the claim that these areas contain a single empty cell. It is a controlled test of alignment-guided distillation at their intersection, with matched controls and participant-aware evaluation used to separate measurement from intervention.

Contributions.

This thesis makes three empirical contributions.

- It establishes controlled measurability and quantifies how ordinary distillation and language-model quality constrain the interpretation of alignment changes [E002][E003][E006][E015].
- It shows that participant averaging and the choice of inference unit alter the supported intervention claim: an averaged-target trend does not establish a mean participant-level benefit [E004][E005][E008][E010][E014].

- It separates four outcomes in dense-proxy distillation. Students learn to predict the TRIBE target, and their retained parameters and representations change while language quality remains comparable. However, the saved text-feature target cannot isolate target content, and the mean participant-level TRIBE-minus-KD estimate on recorded fMRI is near zero [E016][E025][E026].

Brain-relevant information is measurable in the tested language-model representations, but the tested distillation regimes do not demonstrate a reliable benefit from neural-target supervision. This conclusion is restricted to the recorded objectives, models, datasets, controls, and biological endpoints.

2 Related work

The literature relevant to this thesis differs along two main axes: whether brain alignment is measured or optimized, and whether the deployed model is retained or compressed. Table 1 locates work that directly uses a brain-alignment quantity. Non-neural feature and privileged-information distillation provide adjacent precedents [11, 12], so the table is a scope map rather than a novelty claim. The narrower question here is whether a neural target contributes target-specific value to a fixed-budget student and transfers to recorded brain responses that were not optimized during training.

Table 1: Selected direct brain-alignment work by deployed-model intervention and treatment of alignment.

	Alignment measured	Alignment optimized
Base model retained or adapted	frozen-model encoding studies [2, 13, 14]	brain-guided fine-tuning and adaptation [8–10, 15]
Model compressed or replaced by a smaller student	post-compression measurement [4]	alignment-guided distillation (this thesis)

2.1 Measuring brain alignment

Most brain–language-model studies treat a pretrained model as fixed and fit an encoding model from its internal representations to neural responses elicited by the same stimulus. This approach has established that alignment is structured rather than uniform across a network. Middle layers often predict language-selective activity better than early or final layers, while syntactic and semantic contributions vary across cortical regions [2, 5]. Across model families, alignment generally increases with language-model capability and scale before flattening in larger systems [13, 14]. These observations motivate the use of middle-layer representations in the present experiments, but they remain correlational: they describe where a brain score is high after training and do not show that changing the score changes the model.

Encoding results also depend on the level at which the neural target is summarized. Sentence-level ROI averages provide a compact benchmark with relatively high reliability, whereas naturalistic voxelwise datasets preserve temporal and spatial detail at the cost of harder alignment and fewer independent participants. Participant averaging emphasizes stimulus-locked structure shared across people and can improve predictability, but a high group-average score does not imply that the same relationship is strong or homogeneous for individual participants. This distinction appears throughout the broader literature because many benchmark comparisons optimize measurement reliability while intervention claims implicitly concern biological generalization. The present work treats those goals separately rather than using benchmark reliability as a substitute for participant-level inference.

The relationship between model quality and alignment is especially important for intervention studies. Better next-token prediction can accompany stronger encoding performance, so a comparison between models of unequal language quality may attribute an ordinary capability difference to a neural objective. The expanded analysis in this thesis confirms a cross-family association on the standard participant-averaged benchmark while also showing that the relationship is weaker in the capable operating range and difficult to estimate within low-signal individual participants [E015]. Accordingly, language quality is treated as a control variable rather than as evidence that alignment is either reducible to perplexity or independent of it.

2.2 Optimizing neural alignment

Intervention studies add neural-prediction or representation terms during model training. In speech, fMRI objectives have been used to adapt compact encoders, with reported improvements in neural prediction and downstream speech tasks; multi-participant variants report additional data-efficiency gains [8, 16]. Word-aligned ECoG targets provide a higher-temporal-resolution variant that improves held-out same-participant neural prediction relative to pretrained, target-permuted, temporal-mean, and temporally shuffled controls [17]. An external biological-transfer test adapts WavLM on deeply sampled fMRI from three people and reports improved ECoG prediction in nine different patients hearing different material [10]. That result crosses participants, stimulus, and measurement modality, but it fits a fresh ECoG readout and compares primarily with the pretrained model rather than with a target-matched nonbrain intervention.

In text, brain-guided objectives have been applied through full fine-tuning, adapters, auxiliary prediction heads, and representation interventions [9, 15, 18, 19]. Merlin, Moussa, and Toneva [9] compare brain-prediction tuning with language-model tuning on the same stimulus text and report a larger share of probes on which brain-tuned models significantly outperform stimulus-tuned models; joint tuning has the highest aggregate win rate. The brain and stimulus arms share the input text, LoRA rank, learning rate, and broad training duration, but are not matched on target geometry, predictability, loss scale, or optimization difficulty. Xiao, Du, and Lin [15] report that task-fMRI-derived representation directions can improve reasoning through intervention or fine-tuning, but do not train a smaller student or test a matched-quality compression frontier. Matched-quality adversarial interventions also provide evidence that deliberately reducing alignment can impair downstream behavior [20].

These positive studies differ substantially in modality, target construction, trainable component, controls, inference unit, and downstream endpoint. A reported improvement can establish that one neural objective worked under its own design without identifying a general property of “brain guidance.” This thesis uses reduced external reproductions only as bounded checks of local mechanism transfer. Failure to reproduce a gain on one participant, a smaller corpus, or a different implementation is not treated as a refutation of a full multisubject result.

Most brain-guided training studies preserve the base model’s parameter budget and begin from a pretrained representation that is already aligned. Their positive results can therefore combine inherited alignment, fine-tuning drift, language-quality change, target learnability, target geometry, and brain-specific structure. The present experiments probe these explanations with initialization controls, quality matching, target-permuted twins, participant-level inference, a recorded-brain endpoint not optimized during training, and a separate comparator-validity audit.

Related neural-regularization studies in vision and audio illustrate why matched controls matter. Neural supervision can improve performance or robustness, but shuffled-label or non-neural objectives may reproduce part of the benefit [21, 22]. Such results do not make the neural objective uninformative; they show that a gain over an untuned baseline is insufficient to identify which part of the training signal caused it.

2.3 Privileged targets and feature distillation

A neural target in this thesis serves as training-only side information and a dense auxiliary target. This is adjacent to, but not identical with, generalized distillation: the present method directly regresses a neural target through an auxiliary head, whereas generalized distillation has a teacher trained on privileged features produce labels for a student. Its analysis makes any advantage conditional on the teacher and student function classes, their approximation errors, and the ease of student imitation [23].

Wu, Zhu, and Wang [12] use expert annotations or future context as training-only privileged information for smaller dialogue students. Saadi and Wang [11] prioritize teacher directions that affect task output rather than every hidden coordinate. These examples show why target uptake is only a manipulation check: a predictable target can change a student without improving recorded-brain prediction. Neural-content attribution requires a sufficiently matched nonbrain target, followed by evaluation on an endpoint not optimized during training.

2.4 Alignment under compression

Compression studies ask a complementary question: how much alignment remains after a model is made smaller. Oota et al. [4] report that several selected post-training quantization and pruning conditions retain similar raw encoding scores, while other methods and smaller checkpoints degrade. The comparisons do not use an equivalence design, nuisance-subtracted endpoint, or controlled knowledge-distillation intervention. They are evidence about a fixed representation after post-training transformation, not about using neural data to train a student.

Knowledge distillation changes the problem because it constrains a student’s outputs while leaving substantial freedom in its internal representation. A student can match a teacher’s next-token distribution without reproducing the teacher’s middle-layer geometry, and a student trained from scratch also learns from the stimulus corpus along a path not mediated by fixed teacher features. The empirical question is therefore twofold: whether ordinary distillation preserves alignment, and whether an explicit neural term can recover or improve what is lost without merely changing language quality. The present experiments address both parts, including initialization, quality, capacity, loss-form, participant, and synthetic-target variants [E003] [E004] [E008] [E011] [E013] [E016] [E017].

This freedom creates both the opportunity and the identification problem. If several internal representations yield similar output distributions, an auxiliary target could in principle select among them without changing perplexity. If the student is under-trained or initialized differently, however, the same representational difference can arise from ordinary quality loss or inherited pretraining. Cold-start, warm-start, ordinary fine-tuning, and matched-perplexity arms are therefore not interchangeable baselines. They isolate transmission through the teacher channel, drift from a pretrained solution, change caused by more text training, and change remaining after output quality is held approximately fixed.

2.5 Validity, quality matching, and inference units

Brain scores are sensitive to evaluation design. Temporal autocorrelation can leak information across shuffled splits, low-level stimulus properties can support apparently positive prediction, and flexible feature spaces can exceed weaker nuisance baselines without isolating the construct of interest [6, 7]. The same scalar score may also arise from different linguistic or acoustic features across models [5]. For these reasons, a held-out score is interpretable only with explicit splits, nuisance variables, controls, reliability criteria, and an identified inference unit.

Three distinctions govern the interpretation in this thesis. First, contiguous sentence blocks or whole stories prevent adjacent naturalistic samples from appearing on both sides of evaluation. Second, a same-architecture untrained model and a target-permuted training twin answer

different questions: the former tests what language training adds to measurement, whereas the latter tests whether an intervention depends on the correspondence between stimulus and neural target. Third, repeated folds, seeds, voxels, and participants are not interchangeable replicates. Inference over five shared stimulus folds cannot be converted into population evidence by adding seeds, spatially dependent voxels do not create independent participants, and a response averaged across people defines a different estimand from the mean effect across individuals [E004] [E006] [E008] [E014].

An intervention control must also support the attribution being made. A target-permuted twin tests whether stimulus-target correspondence matters, but it does not show that a neural target is better than another structured auxiliary target. Equal output width does not match covariance geometry, effective rank, baseline predictability, nuisance predictability, loss scale, or the amount of induced student change. Matching these properties would not prove information equivalence, but large mismatches can invalidate a content-specific comparison [E026].

Quality matching completes this validity framework. Because alignment co-varies with language-model quality over the broad model range, a neural intervention must be compared with a non-neural alternative at comparable held-out quality whenever the claim is objective-specific [E015]. Matching quality still does not identify the content of an internal representation, but it removes a major alternative explanation for a higher alignment score. The present work tests whether neural-target supervision improves participant-level alignment in a compressed student and audits whether the available nonbrain comparator supports target-specific attribution.

3 Methods

3.1 Study overview

The study follows five gates rather than treating every alignment score as the same kind of evidence. Gate 1 validates measurement: a trained language model must predict held-out brain responses beyond low-level nuisance features and a same-architecture untrained network. Gate 2 measures preservation and quality: ordinary distillation must be characterized before a brain loss is credited with changing anything. Gate 3 tests intervention: training against the real target must outperform an otherwise matched control target. Gate 4 tests biological generalization: the contrast must survive the participant, not a voxel or seed-fold cell, as the independent unit. Gate 5 tests usefulness: a reliable brain-specific representation change must improve an external endpoint at comparable language-model quality.

The experiment families instantiate these gates at increasing levels of difficulty. ROI-scale sentence experiments provide a controlled and inexpensive feasibility setting. Voxelwise naturalistic experiments test whether the measurement survives whole-story splits and subject-specific responses. Participant-level, alternative-loss, full-fine-tuning, cognitive-target, external-reproduction, and synthetic-target experiments probe distinct failure explanations. No single family is treated as a universal test of neural supervision.

3.2 Brain data and response targets

Functional magnetic resonance imaging (fMRI) measures changes in blood oxygenation associated with neural activity. The result at one spatial location, called a *voxel*, is a noisy time series rather than a direct measurement of individual neurons. A *region of interest* (ROI) pools voxels selected for an anatomical or functional role, such as language processing. A *participant* is one person whose responses were recorded; voxels within that participant are repeated spatial measurements, not independent biological replicates.

The blood-oxygen response is delayed and blurred relative to the words that elicited it. For isolated-sentence data, the released response already summarizes the response associated with

each sentence. For continuous stories, word-level model features must be aligned to the scanner sampling grid and accompanied by delayed copies or a hemodynamic response model so that the encoding model compares compatible time points. This temporal step creates a major leakage risk because adjacent scanner samples and adjacent story segments are correlated. Holding out complete contiguous stories or blocks is therefore part of the estimand: it asks whether a mapping learned on one stretch of language predicts a genuinely unseen stretch rather than its temporal neighbors.

An ROI and a voxelwise target answer complementary questions. ROI averaging reduces dimensionality and measurement noise, which makes controlled screens feasible, but it can conceal spatially heterogeneous effects and yields only a handful of target dimensions. Voxelwise evaluation retains spatial detail and can reveal whether a pattern is widespread, but thousands of voxels from one person remain dependent measurements from that person. Neither representation is intrinsically more biologically general. Population generalization requires independent participants regardless of the spatial resolution used inside each participant.

We use two principal fMRI resources. The Tuckute benchmark contains responses to 1,000 isolated sentences in five left-hemisphere functional language ROIs [1]. Its main response is averaged across participants and voxels within each ROI, which improves signal-to-noise ratio but changes the estimand from an individual response to a group-average target. The matched mean noise ceiling for these five ROIs is a correlation-scale value of $r_{\text{NC}} = 0.491$; it is reported as reliability context and is not used as a denominator for a variance-scale unique- R^2 score [E002][E004].

Participant averaging is performed on responses before the encoding model is evaluated. If $Y_i(S) = g(S) + \epsilon_i(S)$, where g is a stimulus-locked component shared across participants and ϵ_i contains participant-specific response and measurement noise, averaging suppresses parts of ϵ_i while retaining g . The resulting target can be more reliable and easier to predict, but it represents the shared group response rather than the expected response of a randomly sampled individual. Because ridge fitting and R^2 are nonlinear operations, scoring an averaged response is not equivalent to averaging participant-specific scores. E005 and E014 use the former estimand, while E008 uses the latter.

The LeBel resource contains naturalistic story-listening fMRI from deeply sampled individuals [24]. The primary signal-validation analysis uses participant UTS03, 20 stories, and whole-story-held-out folds. Voxels are retained only when their split-half reliability exceeds 0.5 on a separate story repeated ten times, leaving 11,442 reliable voxels [E006]. The inference supported by this experiment is conditional on UTS03 and this acquisition and selection protocol. The many retained voxels increase spatial resolution but do not turn one participant into a population sample.

Reliability selection and noise-ceiling normalization are also different operations. Selection uses an independently repeated story to exclude response locations whose repeat measurements are too unstable for meaningful evaluation. Normalization would rescale a performance metric by an estimated maximum on a compatible scale. The thesis performs the former for LeBel and reports the Tuckute correlation ceiling as context, but it does not divide a variance-scale unique R^2 by a correlation-scale ceiling. This distinction preserves the raw experimental contrasts while preventing an apparently intuitive but dimensionally invalid “fraction of ceiling” claim.

The intervention studies use three response constructions. An averaged Tuckute target asks whether training helps predict the common response after participant averaging. Participant-specific Tuckute targets ask whether the effect generalizes across people, with the participant as the independent unit. LeBel voxelwise targets ask whether a dense within-participant regression objective improves held-out alignment under naturalistic evaluation. Because averaging occurs before the nonlinear encoding score is calculated, the averaged and participant-specific analyses estimate different quantities.

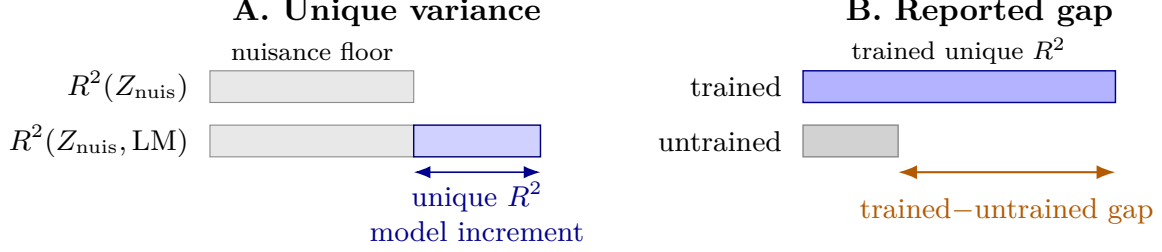


Figure 3: The two alignment contrasts. Unique variance is the held-out R^2 increment obtained by adding language-model features to a nuisance model. The trained-over-untrained gap subtracts the same quantity measured with a randomly initialized network of the same architecture. Bar lengths are illustrative.

3.3 Alignment measurement

The concrete measurement question is whether a frozen model representation predicts brain responses for stimuli that were not used to fit the readout. Let n be the number of stimulus observations, let $X \in \mathbb{R}^{n \times d}$ contain d features from one language-model layer, and let $Y \in \mathbb{R}^{n \times v}$ contain responses from v voxels or ROI targets. The encoding map is fitted by ridge regression:

$$\hat{W} = \arg \min_W \|Y - XW\|_F^2 + \alpha \|W\|_F^2. \quad (1)$$

Here W maps model features to responses, $\|\cdot\|_F$ is the Frobenius norm, and α controls shrinkage. In plain language, the first term rewards accurate brain prediction and the second prevents a high-dimensional readout from fitting noise. The model is linear in the supplied representation, the ridge parameter is selected without using the test fold, and all learned preprocessing is fitted within the training fold. Equation 1 licenses a claim about linearly accessible predictive structure; it does not identify a shared neural mechanism.

A raw encoding score can be positive because the representation carries sentence length, timing, word identity, or other stimulus properties that also affect the brain response. The relevant question is therefore how much predictive value the contextual representation adds beyond a declared nuisance set Z . We estimate this increment with cross-validated semipartial R^2 :

$$\Delta R^2 = R_{\text{cv}}^2(Y; [Z, X]) - R_{\text{cv}}^2(Y; Z). \quad (2)$$

The full model receives nuisance and language-model features, while the reduced model receives the nuisance features alone. Positive ΔR^2 means that X improves held-out linear prediction beyond this particular Z . The claim is conditional on the nuisance set and evaluation design: it does not mean that every stimulus-predictable alternative has been removed.

Naturalistic stimuli create an additional shared-cause problem because even an untrained network can encode tokenization, architecture, and residual stimulus structure. We therefore define the trained-over-untrained gap:

$$\Delta_{\text{train}} = \Delta R_{\text{trained}}^2 - \Delta R_{\text{untrained}}^2. \quad (3)$$

Both arms use the same architecture, nuisance features, preprocessing, and folds. The subtraction asks what language training adds over random initialization under that matched design [E006]. It removes neither every architectural confound nor every possible common cause, so the licensed conclusion remains a trained-over-untrained predictive difference rather than a mechanistic identity.

Noise ceilings quantify response reliability rather than model performance. A correlation ceiling r_{NC} and a variance ceiling r_{NC}^2 are different quantities and cannot be interchanged. We use reliability ceilings to define or contextualize measurable targets, but we do not divide semipartial R^2 by a correlation-scale ceiling.

3.4 Controls, uncertainty, and inference units

All time-series analyses use contiguous held-out units. The ROI benchmark uses contiguous sentence blocks, and the naturalistic analysis holds out complete stories. Randomly shuffling neighboring fMRI time points across train and test would allow temporally autocorrelated observations to leak across the split, so such splits are not used for scientific claims [6].

Nuisance capacity is matched where wide contextual and static feature blocks compete. Principal components are learned on each training fold and reduce those blocks to the same rank, while low-dimensional scalar nuisances remain unprojected. This prevents the contextual block from winning solely because it provides more columns [E006]. Surprisal is not included as a nuisance because it is itself derived from a language model and would remove part of the construct being tested; static lexical frequency remains a nuisance.

Reliability selection is independent of the evaluated stories. The voxelwise analysis uses a separately repeated story to select reliable voxels before cross-validation [24, 25]. This design avoids choosing voxels because they happened to correlate with the evaluated model. It does not establish generalization to other participants or scanners.

Intervention experiments compare a real target with a control target under matched training settings. The block-permuted twin reorders contiguous response blocks while preserving the loss shape and most optimization conditions. It is a specificity control, not a perfect randomization null, because some implemented permutations can leave observations paired with their original targets [E016]. Conclusions therefore use the observed real-minus-control contrast and do not describe the twin as exactly marginal-preserving.

The estimand, estimator, and independent unit are stated separately. For participant-specific intervention, the estimand is the expected participant-level real-minus-control change in held-out ΔR^2 . The estimator first averages paired seed and fold contrasts within each participant and then compares participants. The nine participants are the biological inference units; the 15 seed-fold cells within one target are repeated computational evaluations and cannot be counted as 15 independent people [E005][E008].

Confidence intervals and tests follow the independent unit available to each experiment. Participant-level tests use participants, multi-seed engineering tests use seeds, and story-level signal validation uses held-out story folds while explicitly retaining the shared-training-data dependence caveat. Power and minimum detectable effect calculations are interpreted on the same scale and unit as the corresponding test. They describe what the design could detect; they do not convert a nonsignificant estimate into proof of zero effect.

Three examples make the distinction operational. In E004, three training seeds are evaluated on each of five held-out sentence folds; the five folds determine uncertainty about stimulus-block sensitivity because seeds within a fold share its test data. In E008, seed and fold contrasts are summarized within each person before the nine participants determine biological uncertainty. In E016, six training seeds are evaluated on the same participant-averaged Tuckute endpoint, so the sign test measures repeatability over optimizer randomness for that endpoint and says nothing about variation across people. Reporting the largest available count in each case would answer a different and usually easier question than the scientific claim.

3.5 Knowledge distillation and intervention objectives

Knowledge distillation asks a student to match the teacher’s softened next-token distribution. Let p_T^τ and q_S^τ denote teacher and student token distributions at temperature τ . The standard distillation loss is

$$\mathcal{L}_{\text{KD}} = \tau^2 D_{\text{KL}}(p_T^\tau \| q_S^\tau). \quad (4)$$

The KL divergence penalizes token distributions that differ from the teacher, and τ exposes relative probabilities beyond the most likely token. Equation 4 constrains the student’s output

distribution. It does not uniquely determine the student’s middle-layer geometry because many internal representations can produce similar logits.

Perplexity is the exponential of held-out mean token cross-entropy. On a shared evaluation corpus, lower perplexity means a better next-token distribution under this metric. Matched compute compares arms at the same nominal training budget, whereas matched perplexity compares them at similar achieved language-model quality. Both are reported because equal steps need not produce equal quality, and an alignment difference caused by under-training is not an objective-specific brain effect.

The primary intervention adds a brain-target regression term:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{KD}} + \lambda_B \mathcal{L}_B. \quad (5)$$

Here $\mathcal{L}_B$ is either mean-squared error to a recorded or predicted brain target or a declared contrastive alternative, and λ_B is its weight. For a mean-squared-error (MSE) target, a learned linear prediction head maps the selected student hidden state into the target space, such as five ROI values or a dense cortical-response vector. The prediction head and the permitted student parameters are optimized jointly, but only the student is retained as the language model after training; the head is used to impose the auxiliary geometry and to check target fit. The block-permuted twin uses the same head architecture, trainable parameters, and optimizer while changing which target is paired with each stimulus. The equation asks whether the auxiliary target selects a student representation that ordinary distillation would not select. It assumes only that the two losses can be optimized jointly. It does not imply that $\mathcal{L}_B$ measures cognition, that increasing λ_B increases alignment, or that an easier target transfers to real brain responses.

Cold-start students are initialized randomly, so any retained alignment must be learned during training. Warm-start students begin from pretrained weights and measure drift under further optimization. Low-rank adaptation (LoRA) interventions restrict updates to small trainable adapter matrices, while full fine-tuning permits all model parameters to change. The experiments vary adapter rank, brain-loss weight, loss form, and initialization to distinguish insufficient optimization capacity from absence of a specific effect.

Teacher, student, corpus, batch construction, and evaluation are held fixed within each paired real-versus-control contrast. The alignment layer is declared before scoring. The early preservation study includes a known compute caveat: the cold arm receives about twice the optimizer steps of the warm arms, so its cross-arm result is interpreted together with held-out perplexity rather than as a pure initialization effect [E003].

3.6 Formal interpretation and its limits

3.6.1 Conditional dependence and unique variance

The formal question is what dependence remains between representation X and response Y after nuisance variable Z is known. Conditional mutual information defines that population quantity:

$$I(X; Y \mid Z) = H(Y \mid Z) - H(Y \mid X, Z). \quad (6)$$

$H(Y \mid Z)$ is the uncertainty in the response after knowing the nuisance, and $H(Y \mid X, Z)$ is the remaining uncertainty after also knowing the representation [26]. Thus Equation 6 is zero when X adds no conditional dependence beyond Z . It licenses a statement about a fully specified joint distribution, not automatically about a finite cross-validation estimator.

Under a population linear-Gaussian model with correctly specified, unregularized conditional means, conditional mutual information is related to partial R^2 by

$$I(X; Y \mid Z) = -\frac{1}{2} \log(1 - R_{\text{partial}}^2), \quad \Delta R_{\text{semi}}^2 = (1 - R_{\text{reduced}}^2) R_{\text{partial}}^2. \quad (7)$$

R_{partial}^2 rescales the improvement by variance left unexplained by the reduced model, whereas ΔR_{semi}^2 is the raw improvement in total variance explained. The equations show why both quantities express incremental dependence under the stated population assumptions. Our estimator in Equation 2 is instead finite-sample, cross-validated, and ridge-regularized. It is therefore an operational measure of incremental linear predictive value, not an estimator or lower bound for conditional mutual information.

3.6.2 Regularization and a conditional MAP reading

The intervention objective can sometimes be read as regularized maximum likelihood. If $\exp[-\lambda_B \mathcal{L}_B(\theta)]$ is normalizable over parameters and does not depend on the observed text sample, then it defines a prior

$$p_B(\theta) = \frac{1}{C} \exp[-\lambda_B \mathcal{L}_B(\theta)], \quad (8)$$

where C is the normalization constant and θ denotes student parameters. Under those assumptions, minimizing text negative log-likelihood plus the penalty is equivalent to a maximum-a-posteriori estimate. The plain-language interpretation is that the penalty prefers some parameter settings among those that fit the text. In the implemented co-trained-target experiments, however, the brain target and readout are data-dependent, so “auxiliary loss” or “regularizer” is the primary description. Equation 8 does not prove that the preference is biologically correct or beneficial.

3.6.3 Data processing

The data-processing inequality answers a narrow preservation question when the required Markov chain is valid. If a student representation C is constructed only by post-processing a teacher representation T , and brain response B has no other path to C once T is known, then $B \rightarrow T \rightarrow C$ implies

$$I(B; C) \leq I(B; T). \quad (9)$$

The inequality says that deterministic or stochastic post-processing cannot create information about B that was absent from T [26]. Ordinary distillation does not automatically satisfy this chain because teacher and student both receive stimulus or corpus information through side paths. Likewise, a chain such as $\theta^* \rightarrow S \rightarrow B$ is an explicit conditional-independence model, not proof that brain responses contain no information beyond text. Equation 9 motivates an empirical preservation test but neither orders teacher and student alignment scores in ordinary distillation nor supplies a supervision budget.

3.6.4 Generalization information bound

The information-theoretic generalization bound asks how strongly learned parameters depend on the particular training sample. For a learning algorithm viewed as a channel from Z^n to learned parameters W , and a σ -sub-Gaussian loss, the expected generalization gap obeys

$$\overline{\text{gen}} \leq \sqrt{\frac{2\sigma^2 I(W; Z^n)}{n}}. \quad (10)$$

n is the sample size and $I(W; Z^n)$ measures dependence between the output of training and the sampled dataset [27]. The right-hand side increases, rather than decreases, with this mutual information. The bound therefore does not show that adding brain information improves generalization and does not predict a benefit of order $\sqrt{\Delta I/n}$. A brain regularizer could tighten the bound only if it reduces dependence on sampled text while retaining empirical fit, which is an empirical possibility rather than a theorem.

3.6.5 Rate–distortion-style trade-offs

Shannon’s rate–distortion function asks for the minimum information rate needed to represent a source within an expected distortion limit:

$$R(D) = \min_{P_{\hat{S}|S}: \mathbb{E}[d(S, \hat{S})] \leq D} I(S; \hat{S}). \quad (11)$$

S is the source, $\hat{S}$ its reconstruction, d a declared distortion measure, and D the allowed expected distortion [26]. The thesis does not identify model parameters or FLOPs with $I(S; \hat{S})$, and it does not establish brain loss as a Shannon distortion. We use “rate–distortion-style” only as an engineering analogy for the empirical Pareto frontier among model budget, perplexity, and alignment. Equation 11 organizes the trade-off but does not predict a favorable frontier shift.

3.7 Experiment families

Signal validation. The ROI feasibility study compares trained and untrained GPT-2-family and Qwen representations under contiguous sentence folds [E002]. The conditional voxelwise confirmation uses UTS03, whole-story-held-out folds, fold-only feature reduction, a capacity-matched nuisance set, and independently selected reliable voxels [E006]. The first establishes inexpensive feasibility; the second is the primary within-participant signal test.

Quality and preservation. Cold and warm distillation arms measure whether ordinary compression leaves alignment headroom and how alignment covaries with held-out perplexity [E003]. A separate comparison across 22 models from six families uses bits per byte as a tokenizer-comparable quality measure and tests the global association, a capable-model band, and family-cluster sensitivities [E015]. These designs distinguish a broad quality relationship from an objective-specific preservation effect.

Intervention and participant generalization. The first intervention compares real and block-permuted averaged ROI targets across paired seeds and contiguous folds [E004][E005]. The participant-specific extension trains and evaluates targets separately for nine participants, making participant the primary inference unit [E008]. Nested and randomized participant-subset analyses test whether apparent improvement with response averaging is explained by increasing sample size or by the identities of included participants [E010]. A separate encoding-average analysis measures legitimate signal-to-noise improvement from averaging without treating that endpoint as individual-level generalization [E014].

Alternative parameterizations. Higher-rank LoRA tests whether the primary intervention was limited by adapter capacity [E011]. Contrastive ROI losses test whether the mean-squared-error target was the wrong geometry [E013]. A naturalistic voxelwise weight sweep tests whether the regression objective improves held-out alignment at dense spatial resolution, while gentle full fine-tuning tests whether updating all parameters changes the conclusion [E013][E017]. Each experiment addresses one mechanism boundary; none alone licenses a universal substrate null.

Practical and privileged-information routes. The downstream test requires a reliable brain-specific representation fulcrum before estimating out-of-domain perplexity over eight training seeds [E009]. The learning-using-privileged-information route uses gaze-derived training-only information, but gates the intervention on label predictiveness without leakage and stable paragraph-disjoint learning curves [E024]. A gate failure stops the proposed intervention rather than being reported as a completed negative training experiment.

Stimulus-predictable and cognitive targets. A stimulus-predictability analysis conditions on estimated response predictable from the text to ask whether a residual alignment score remains [E020]. Because the reference predictor is itself estimated and can leak or be unreliable, this experiment diagnoses a candidate ceiling rather than identifying a universal ceiling. Reading-time targets test whether a reported cognitive advantage depends on meaningful target content or can be reproduced by content-free signal shape [E021]. Reduced external reproductions of text and speech interventions are treated as corroboration or bounded non-reproduction under the implemented design, not as faithful refutations of the original studies [E019][E022].

Synthetic targets and real-brain transfer. The dense-target experiment distills GPT-2-medium into GPT-2 using 95,999 WikiText training sentences and 1,999 held-out sentences across six seeds [E016]. TRIBE v2 is a pretrained multimodal encoding model that maps naturalistic stimuli to predicted human fMRI responses on a standardized cortical surface [28]. For the text-only E016 stimuli, its output is a 20,484-dimensional vector of predicted cortical locations and is treated as a synthetic neural target rather than recorded fMRI. The student layer 6 hidden state is mapped to that target by the jointly trained prediction head. The textfeat control instead Gaussian-projects a frozen teacher hidden representation to the same nominal width, then uses an otherwise matched prediction head and training path. Dimension matching controls output width but not effective rank, geometry, or information, so held-out target R^2 is interpreted within target family rather than compared as a shared scale.

Saved students pass through a separate transfer gate on 1,000 Tuckute sentences using five contiguous nuisance-subtracted folds and student layer 7 [E016]. For each seed, the primary contrast subtracts the textfeat-over-KD real-brain gain from the TRIBE-over-KD real-brain gain. The inference unit is six training seeds evaluated on one fixed participant-averaged endpoint. The exact sign test therefore supports a statement about repeatability across training seeds for that endpoint, not participant- or population-level biological generalization.

E025 reuses the saved E016 students without retraining and replaces the participant-averaged transfer endpoint with the nine complete Tuckute participants, excluding UID 853 under the established incomplete-ROI rule [E025]. Layer 7 is primary and layer 6 is a mechanistic sensitivity; each participant, model, and seed is scored with the same five contiguous folds, rank-50 fold-fitted nuisance/static/contextual decomposition, and fixed language ROIs [E025]. Seeds are crossed technical realizations and are averaged before participant inference, leaving the participant as the sole biological inference unit. The predeclared internal continuation threshold is $+0.002$ unique R^2 [E025]. It determines whether the positive-mechanism program continues and is not treated as a universal minimum biological effect. The activation rule is conjunctive. It checks participant and seed signs; leave-one-participant and leave-one-block stability; held-out-subgroup behavior; direct TRIBE-over-KD; the result after dropping the participant with the highest KD alignment; and sensitivity to adding imageability to the nuisance set [E025]. Direct bits per byte, retained target learning, and deterministic representation movement are checked before participant inference; failure of any manipulation gate makes the endpoint descriptive rather than evidential.

E026 retrospectively audits whether the saved TRIBE and projected text-feature targets support a content-specific cross-target comparison [E026]. It freezes two disjoint deterministic samples of 4,096 WikiText training items, reconstructs the exact training standardization, and compares centered covariance spectra, stable rank, cumulative variance mass, within-document order structure, KD baseline predictability, remaining headroom, and held-out prediction from low-level and frozen static-text nuisance features. For the already trained six-seed students, it compares global and blockwise parameter displacement from the common KD checkpoints and centered layer-6 representation movement on the fixed held-out WikiText cache. Geometry samples are deterministic checks without population confidence intervals; seeds are technical training realizations summarized by paired values, descriptive t intervals, and exact magnitude-

preserving sign-flip tests. The predeclared margins are engineering identification criteria rather than formal equivalence tests, and initial-gradient comparability remains unknown because those gradients were not retained.

4 Results

The experiments test a sequence of increasingly demanding claims. First, an alignment score must detect information that language training adds to a model representation. Second, ordinary distillation must leave enough alignment headroom for an additional objective to matter. Third, an alignment-guided objective must move held-out alignment beyond an appropriate control. Fourth, that movement must generalize across participants or another biological unit. Finally, it must improve an outcome that was not optimized during training.

The measurements in this sequence do not share an inference unit. For compactness, let A denote held-out nuisance-subtracted unique R^2 , and let u , s , and k index participant, training seed, and held-out stimulus block. Table 2 states the primary estimand and the order of aggregation for each claim.

Table 2: Estimand map for the experimental chain. Each estimator aggregates repeated technical measurements before using the unit and scope named in the third column.

Claim	Estimand and estimator	Unit and inferential scope	Result
Controlled measurement	Mean held-out $A(\text{trained}) - A(\text{untrained})$ across fixed contiguous blocks.	Held-out blocks within each fixed target; fitting folds share training items, and no population inference is made.	Positive on both recorded-fMRI substrates [E002][E006].
Recorded-target intervention	Average $A(\text{real}) - A(\text{control})$ over seeds and folds within each participant, then average participants.	Nine participants after seed and fold aggregation.	No supported participant-level benefit in the tested regimes [E008][E011][E013].
Dense-proxy uptake	Seed-paired target- R^2 gain over KD, plus parameter and representation change from KD.	Six technical training seeds for one fixed procedure; no biological or target-population inference.	Target fit and student movement pass [E016][E025].
Comparator identification	Frozen conjunction over target geometry, baseline predictability, and seed-paired student movement.	Two deterministic item samples for geometry and six technical seeds for movement; no population inference.	Saved text-feature comparator fails measured-axis comparability [E026].
Biological transfer	Average TRIBE-minus-KD within each participant after seed and fold aggregation; retain the preregistered difference-in-differences as a separate relative comparison.	Nine participants after seed and fold aggregation.	Direct mean is near zero; the relative upper bound is below the continuation threshold but cannot identify target content after E026 [E025][E026].
External utility	Out-of-domain or sample-efficiency contrast, conditional on a verified intervention or informative privileged signal.	Eight seeds for E009; paragraph groups for the E024 precondition.	No payoff claim is licensed [E009][E024].

4.1 Controlled measurability on real fMRI

The first question is whether a trained language model contains brain-predictive information beyond the recorded low-level properties of the stimulus. E002 measures this on the Tuckute sentence benchmark by fitting ridge encoding models from a fixed LM layer to five participant-averaged left-hemisphere language ROIs. The nuisance-only model contains sentence length, position, and static-embedding features, and the full model adds the contextual LM representation at matched PCA capacity. The estimand is the held-out semipartial increment, or unique R^2 , under five contiguous stimulus folds. A same-architecture untrained network controls for feature dimension and architecture without language training [E002].

All three trained models have positive unique R^2 , whereas every untrained seed is negative (Table 3). The trained-minus-untrained gaps are +0.030 for GPT-2, +0.033 for GPT-2-medium, and +0.050 for Qwen2.5-0.5B [E002]. The corrected mean ceiling coefficient for the five functional ROIs is 0.491 [E002][E004]. Because that coefficient and unique R^2 are not the same statistical quantity, their legacy ratio is not interpreted as a fraction of explainable variance. These

estimates belong to one participant-averaged target evaluated over five stimulus folds; they establish conditional measurability on this benchmark, not population prevalence.

Table 3: Tuckute ROI benchmark under five contiguous folds. Unique R^2 is the held-out semipartial increment over the nuisance set [E002].

Model	Layer	Trained ΔR^2 (mean $\pm$ SD)	Untrained ΔR^2	Gap	Source
GPT-2	L7	$+0.020 \pm 0.008$	-0.011	$+0.030$	[E002]
GPT-2-medium	L14	$+0.019 \pm 0.005$	-0.014	$+0.033$	[E002]
Qwen2.5-0.5B	L12	$+0.036 \pm 0.010$	-0.014	$+0.050$	[E002]

E006 asks whether the same trained–untrained contrast survives at voxel resolution in naturalistic data. It uses the 20-story LeBel data for participant UTS03 and selects 11,442 voxels using split-half reliability from a separate repeated story. The encoding model holds out entire stories and subtracts word and phoneme rate, duration and length, log frequency, and static-embedding nuisance features. The primary estimand is again the trained-minus-untrained unique- R^2 gap on the fixed UTS03 endpoint [E006].

The gap is $+0.0207$ for GPT-2 and $+0.0277$ for Qwen2.5-0.5B. Ninety-five percent and 99% of selected voxels, respectively, have positive pointwise gaps, and each model has a positive mean gap on all five held-out story blocks [E006]. The untrained unique R^2 is -0.017 in both model comparisons [E006]. These spatial and story-block signs show that the result is not confined to a small voxel set or one held-out story. They are not independent replicates: all voxels come from one participant, voxels are spatially dependent, and the story folds share training stories. The original voxel-bootstrap confidence intervals and the E006-derived E007 minimum-detectable-effect claim are therefore retracted [E006].

Table 4: Conditional voxelwise result on LeBel UTS03. Spatial prevalence and story-block signs are sensitivity checks within one participant, not population inference [E006].

Model	Layer	Trained–untrained gap	Positive selected voxels	Positive story blocks
GPT-2	L7	$+0.0207$	95%	5/5
Qwen2.5-0.5B	L12	$+0.0277$	99%	5/5

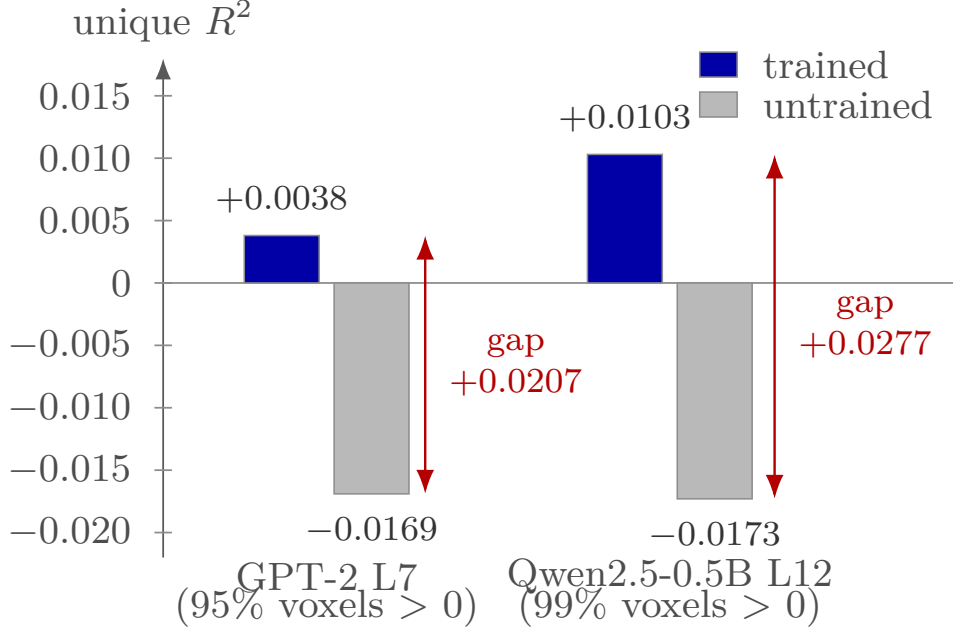


Figure 4: Mean unique R^2 on the selected LeBel UTS03 voxels. Arrows show the trained-minus-untrained gaps, and labels report spatial prevalence within UTS03 [E006].

E002 and E006 therefore establish the first gate: language training adds linearly accessible information that predicts held-out fMRI beyond the recorded nuisance set. They do not show that the information can be increased by training, that it generalizes across participants, or that increasing it would improve language modeling. The two datasets contribute different parts of that statement. Tuckute shows the effect on isolated sentences and a compact participant-averaged language-network target across three model families, while LeBel shows it on continuous stories with temporal controls and a dense response from one deeply sampled participant. The common trained-over-untrained direction makes the measurement premise less dependent on a single stimulus format. The different biological units prevent their combination into one population effect, so they are treated as complementary existence evidence rather than a meta-analysis.

4.2 Distillation leaves headroom, but alignment is coupled to model quality

E003 asks whether ordinary logit knowledge distillation preserves the teacher’s alignment without an explicit brain objective. The primary arm initializes a GPT-2 student randomly and trains it from the logits of GPT-2-medium. Warm-start KD, ordinary language-model fine-tuning, pretrained GPT-2, distilgpt2, and an untrained network provide reference points. All arms are scored at fixed layers on the same participant-averaged Tuckute target and nuisance-controlled contiguous folds [E003].

The cold-init student ends +0.0181 below its teacher in unique R^2 ($p < 0.001$ under the recorded bootstrap) and retains 0.37 [0.14, 0.58] of the teacher–floor interval [E003]. This rules out preservation for free in the tested from-scratch run and creates measurable headroom. The magnitude is not an objective-specific estimate because the cold student remains under-trained: held-out perplexity is 477, compared with 74 for its teacher and 44–95 for the converged warm arms. Absolute unique- R^2 values also vary with PCA rank, although the retention ordering survives $n_{\text{PCA}} \in \{25, 50, 100\}$ [E003].

Table 5: Distillation and reference arms on the participant-averaged Tuckute benchmark [E003]. Floor-anchored retention is $\rho' = (A_S - A_0)/(A_T - A_0)$; distilgpt2 uses GPT-2 as its lineage teacher.

Arm	Initialization/objective	ΔR^2	ρ' [bootstrap CI]	Held-out ppl	Interpretation
GPT-2-medium	pretrained teacher	+0.0188	not defined	74	teacher reference [E003]
GPT-2	pretrained	+0.0195	1.02 [0.77, 1.27]	105	inherited alignment [E003]
LMFT-warm	warm, language loss	+0.0174	0.95 [0.81, 1.13]	44	drift control [E003]
KD-warm	warm, logit KD	+0.0144	0.84 [0.67, 1.04]	95	drift under KD [E003]
distilgpt2	released distilled model	+0.0076	0.60 [0.42, 0.84]	169	lineage reference [E003]
KD-cold	random, logit KD	+0.0005	0.37 [0.14, 0.58]	477	under-trained headroom
untrained	random floor	-0.0102	0	$\sim 6 \times 10^4$	floor [E003]

The same run exposes the quality confound. Across its five non-teacher trained/reference arms, alignment correlates with log perplexity at Pearson -0.88 [E003]. That correlation uses few heterogeneous points and is not a general law, so E015 repeats the analysis with a tokenizer-independent quality measure across 22 models from six decoder families. Its primary full-range association between mid-layer unique R^2 and log bits per byte is $r = -0.783$, with family-cluster bootstrap 95% interval $[-0.911, -0.500]$ [E015]. Leave-one-family-out estimates remain between -0.738 and -0.833 [E015]. The 22 models are not 22 independent scientific replicates: the six families form approximately two model generations, and models within scaling families share training recipes.

The coupling weakens where current brain-tuning comparisons operate. For models with bits per byte below 1.30, $r = -0.501$ with Fisher interval $[-0.859, +0.188]$; in the overlap band $[1.13, 1.26]$, $r = -0.480$ with interval $[-0.885, +0.340]$ [E015]. Those capable-band intervals include zero. A scale-adjusted partial correlation remains negative at -0.675 with Fisher interval $[-0.857, -0.344]$, but bits per byte and parameter count are highly collinear, so this does not isolate a pure quality mechanism [E015]. Family residuals are also not identified because the families lack common quality support; the recorded ANCOVA family test has $p = 0.204$ [E015].

The combined conclusion is narrower than either “KD destroys brain information” or “alignment is just perplexity.” Ordinary KD can leave a student below its teacher, but E003 does not distinguish objective-specific shedding from under-training and quality. E015 shows that quality is a necessary matching variable across a broad modern-decoder comparison, while its capable-band and architecture-residual analyses show that output quality does not determine the hidden representation. The intervention experiments therefore compare real brain targets with controls matched on training procedure or perplexity rather than with a vanilla model alone. This distinction also changes the interpretation of “headroom.” The teacher-student gap is an opportunity only if some of it is caused by an internal choice that an auxiliary objective can alter without paying the same cost in perplexity. E003 establishes the gap but not that counterfactual. E015 makes unequal quality a credible alternative explanation, yet its capable-band uncertainty leaves room for representations of similar output quality to differ in alignment. The intervention stage is needed precisely because neither preservation nor cross-model correlation identifies the movable component.

4.3 The apparent intervention gain depends on the inference target

E004 is the first direct intervention test. It co-trains a Qwen2.5-0.5B student with an MSE brain term and compares it with an otherwise identical block-permuted-target twin [E004]. Stimulus folds, optimization budget, nuisance model, student, teacher, and brain-loss weight are held fixed. The estimand is the real-minus-permuted change in held-out unique R^2 relative to the same untuned base [E004].

The point estimate is $+0.00316$. The original interval treated three seeds by five folds as 15 independent observations and excluded zero. After averaging seeds within fold, the valid five-

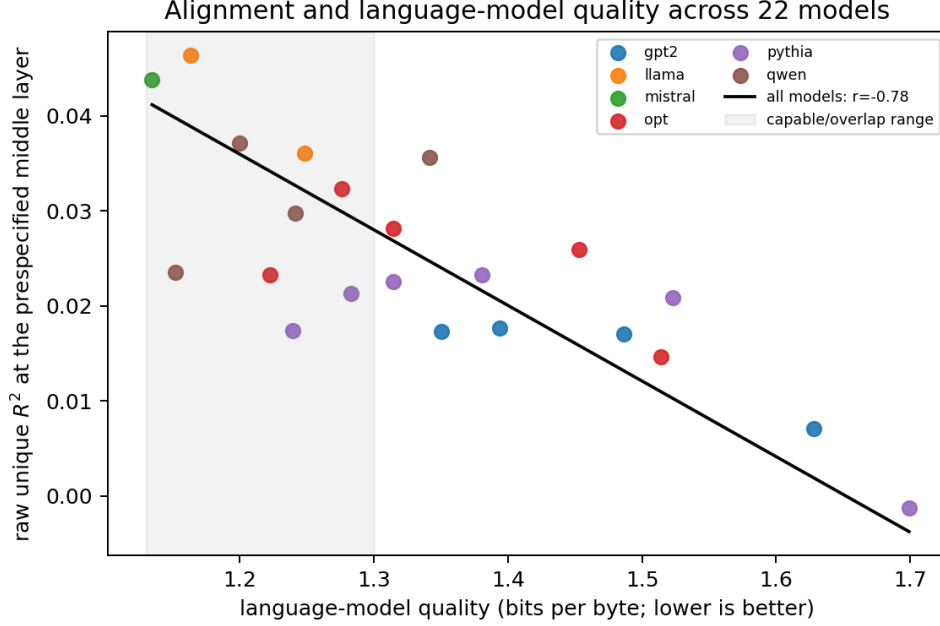


Figure 5: Tokenizer-independent language-model quality and controlled mid-layer alignment across the 22 E015 models. The fitted line summarizes the full range; the shaded region marks the capable/overlap range in which the association is weaker and individually uncertain. Points share family and generation structure, so they are not 22 independent scientific replications [E015].

fold t interval is $[-0.00227, +0.00858]$ [E004]. Fold 4 supplies 64% of the summed contrast, and removing it reduces the estimate to $+0.0014$. The run’s own unpaired permutation exceedance flag is false. E004 therefore provides no inference-unit-valid evidence that the objective improves held-out alignment, while its five folds cannot exclude a small effect near the observed mean.

E005 asks the related matched-perplexity question against the five-participant-averaged target. At $\lambda = 10$, its real-minus-permuted mean is $+0.0081$. The flat seed-by-fold bootstrap interval $[+0.0023, +0.0171]$ is pseudo-replicated; the five-fold t interval is $[-0.0030, +0.0193]$, the median is $+0.0034$, and one fold supplies 52% of the mean [E005]. Four of five fold means are positive, so the result is a small averaged-target trend rather than a precise null.

The later λ sweep asks whether alignment can improve without worsening language quality or whether any gain requires a language-quality cost. At $\lambda = 1$, the real-minus-matched-permuted contrast is $+0.0030$ with fold interval $[-0.0025, +0.0085]$ [E005]. At $\lambda = 3$, it is $+0.0063$ with interval $[-0.0092, +0.0218]$, and at $\lambda = 10$ it is $+0.0072$ with interval $[-0.0044, +0.0188]$ [E005]. At $\lambda = 30$, the contrast is $+0.0060$ $[+0.0002, +0.0119]$, but held-out perplexity worsens from approximately 51.5 for KD-only to 67.4 [E005]. The only fold-level-positive arm therefore buys alignment by moving to a worse language-model operating point, not by improving alignment without a language-quality cost.

E008 changes the biological inference unit. It repeats the real-versus-permuted comparison separately for nine participants, using three optimization seeds and five shared folds within each participant. The estimand is the across-participant mean of each participant’s seed-and-fold-aggregated real-minus-permuted effect. The mean is $+0.00010$, the nine-participant t interval is $[-0.00037, +0.00058]$, and 5/9 participant effects are positive (two-sided sign $p = 0.50$) [E008]. The fold-clustered sensitivity estimate is $+0.00026$ with interval $[-0.0004, +0.0009]$, and every leave-one-fold-out analysis fails to establish a positive effect. The observed participant variance gives an 80%-power minimum detectable effect of approximately $+0.0006$ to $+0.0013$, well below the averaged-target estimate [E008].

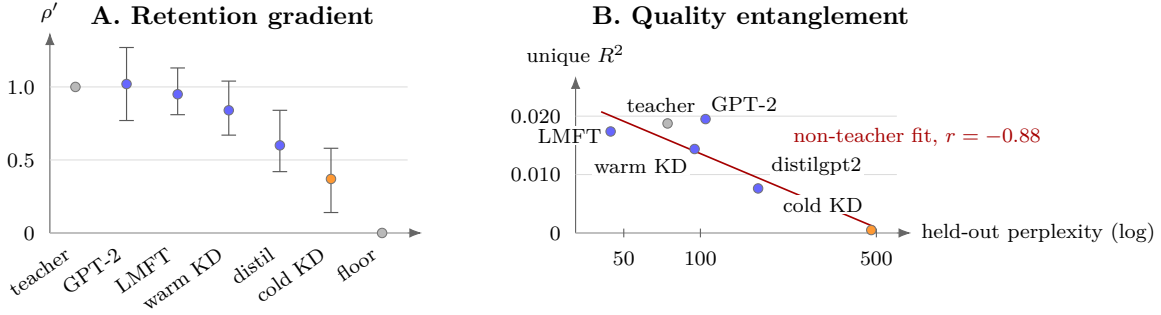


Figure 6: Plain KD leaves alignment headroom, but alignment co-varies with language-model quality in the E003 reference set. The line and coefficient describe the five non-teacher E003 points, not the corrected cross-family E015 law [E003].

The four participants used to construct the earlier averaged target have an individual mean of -0.00010 [E008]. The five held-out participants have mean $+0.00026$, also with an interval containing zero [E008]. Thus, the E005 estimate is not an estimate of a mean per-person effect. It is an effect on a different, higher-SNR target: the response shared across the averaged participants.

E010 tests whether participant count alone causes a smooth increase in the intervention gap. Its initial nested subsets give gaps of -0.0002 , -0.0001 , $+0.0023$, $+0.0194$, and $+0.0071$ for $k = 1, 2, 3, 5, 9$ averaged participants [E010]. Because subject identity changes with k , this design confounds count with which participants enter the average. The random-subset rerun gives $+0.0015$, -0.0013 , $+0.0088$, and $+0.0094$ for $k = 1, 3, 5, 9$, with standard deviations between 0.005 and 0.015 [E010]. The non-monotone, noisy rerun does not identify a causal dose-response law. It remains suggestive corroboration that averaged and individual targets behave differently.

E014 isolates the measurement effect without training an LM. Using the same Qwen representation, it compares encoding of individual targets with encoding of random participant averages. Mean per-participant unique R^2 is $+0.0069$, with 7/9 participants positive, whereas the standard five-participant average scores $+0.0357$ [E014]. The count sequence for $k = 1, 2, 3, 5, 9$ is $+0.0039$, $+0.0162$, $+0.0324$, $+0.0444$, $+0.0702$ [E014]. Because individual scores are already positive, this is not evidence that averaging fabricates an otherwise absent encoding signal. It shows legitimate SNR improvement and a change from individual-response alignment to alignment with the shared response component.

Taken together, E004 and E005 provide small averaged-target trends but no reliable intervention effect at comparable language quality under five-fold inference. E008 supplies the primary biological result: no demonstrated mean per-individual gain in nine participants under the tested participant-specific objective and its same-procedure control. E010 does not convert the averaged-versus-individual contrast into a clean causal dose response, and E014 shows why a larger score on an averaged target can still be a legitimate measurement of a different estimand. The apparent contradiction is therefore resolved at the level of the question being estimated. E005 asks whether training helps predict one denoised group-average response, and its small positive trend is compatible with an intervention that follows highly shared stimulus structure. E008 asks whether the same real-versus-control advantage is positive on average across individual people, and it finds a tightly bounded mean near zero. Neither result invalidates the other measurement, but only the latter supports inference to participant variation. This is why response averaging is described as an estimand change rather than as a universally invalid artifact.

4.4 Alternative intervention regimes do not establish a reliable gain

E011 asks whether the E008 result is caused by insufficient low-rank adaptation (LoRA) capacity. It increases adapter rank from 16 to 64 and doubles the training duration while holding the target, objective, participant set, and same-procedure control fixed. Across the same nine participant units, the mean real-minus-permuted effect is $+0.00043$ with t interval $[-0.0005, +0.0013]$ [E011]. Six of nine effects are positive, but one participant contributes 79% of their sum; leaving that participant out returns the estimate to approximately $+0.00010$ [E011]. Absolute alignment and perplexity move about as much as in E008, so this result establishes robustness to extra adapter capacity at the same effective operating point, not robustness to every stronger manipulation.

E013b changes the loss geometry from MSE to symmetric contrastive training while keeping the nine-participant ROI design. Its mean is -0.00019 with participant t interval $[-0.0008, +0.0005]$, 4/9 positive signs, and a matched-perplexity intercept of approximately -0.0003 [E013]. The contrastive objective therefore does not rescue the ROI result. The scope is limited by the five-dimensional ROI target, which is a weaker test of a contrastive objective than a high-dimensional voxel target.

The voxelwise E013 route tests whether the MSE distillation mechanism can first create an improving representation on naturalistic LeBel data. Without a KD anchor, real-target tuning reduces held-out alignment below the untuned base in all six subject-by-seed runs, so the first run is a failed manipulation rather than a null effect [E013]. With a KD anchor on UTS03, the untuned base is approximately 0.0187 unique R^2 [E013]. At $\lambda = 1$, real-target tuning reaches 0.0183 with real-minus-permuted gap -0.0009 ; at $\lambda = 3$ and 6, it falls to 0.0086 and 0.0130, with gaps -0.0049 and -0.0039 [E013]. No tested weight improves over the base or beats its permuted twin. This is a single-participant mechanism failure for voxelwise MSE distillation, not a population null for every voxelwise objective.

E017 changes the parameterization to full fine-tuning. An aggressive configuration produces no manipulation success in three UTS03 seeds and increases perplexity by a factor of 1.63 [E017]. A gentler configuration preserves perplexity more closely and is then run for three seeds on each of UTS01, UTS02, and UTS03. Across the nine subject-by-seed cells, the recorded real-minus-permuted mean is $+0.0003$, with descriptive cell-level t interval $[-0.0002, +0.0008]$ and $p = 0.274$ [E017]. Per-subject means are $+0.0005$, -0.0001 , and $+0.0006$ [E017]. The valid biological unit is the participant, so the nine-cell interval is not population inference; the run is a three-participant feasibility result showing no demonstrated brain-specific gain in this configuration.

These variants close specific explanations rather than the entire hypothesis space. The ROI result is not rescued by more adapter capacity or a contrastive loss. The tested voxelwise MSE mechanism fails to improve the base representation, and the tested full-fine-tuning configurations either damage language quality or remain indistinguishable from their permuted twins. Different data, targets, objectives, or substantially larger participant cohorts remain outside these results. The sequence nevertheless removes several immediate rescue explanations. If the primary null were caused only by insufficient low-rank capacity, E011 should have enlarged the contrast; if squared-error geometry were uniquely unsuitable for ROI targets, E013b should have improved it; if adapters were the bottleneck, the gentle full-fine-tuning regime in E017 should have separated real and control targets. None does so under its valid scope. The voxelwise branch is even earlier in the causal chain because stronger weights fail the manipulation check itself. These results therefore narrow the plausible success regime toward different data or target construction rather than a simple increase of the same optimization knob.

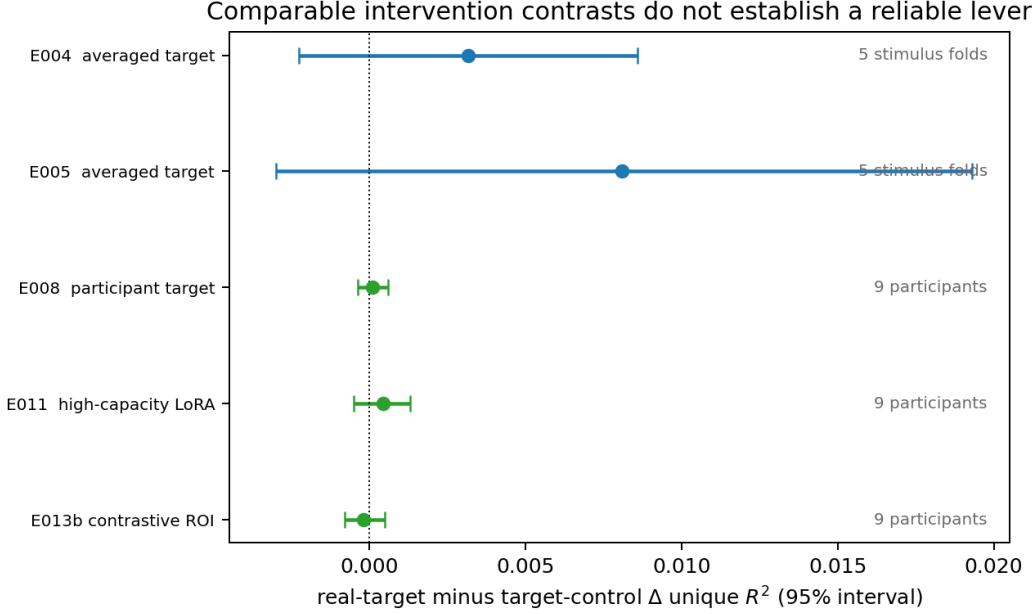


Figure 7: Comparable real-target minus target-control intervention contrasts. E004 and E005 use five stimulus folds; E008, E011, and E013b use nine participants. The rows are displayed together to show scale and uncertainty but are not pooled because their estimands and inference units differ [E004,E005,E008,E011,E013].

4.5 No practical payoff is demonstrated when the prerequisite signal does not move

E009 asks whether alignment-guided KD improves a non-brain outcome: out-of-domain perplexity on Pereira and LeBel text. It compares KD-only, real-brain mean-squared error (MSE), block-permuted MSE, and a text-feature auxiliary target over eight training seeds. The mediator that would make the downstream test interpretable is the real-minus-permuted alignment change on the in-domain averaged target [E009].

That mediator has mean +0.011, median +0.0042, and 5/8 positive seed signs. Its measured 80%-power minimum detectable effect is 0.023, and seed 0 contributes +0.063 while the remaining seven seeds average approximately +0.004 [E009]. The representation fulcrum is therefore not reliable. Every OOD-perplexity contrast also lies inside its measured minimum detectable effect: real MSE minus KD-only is +0.009 on Pereira and +0.053 on LeBel, while real MSE minus the permuted twin is -0.019 and -0.068 [E009]. E009 supports no detected brain-specific out-of-domain (OOD) perplexity payoff under this all-data, averaged-target setup. It is a bounded result because the manipulation whose payoff was being tested did not reliably take hold, and because robustness and sample-efficiency outcomes were not run.

E024 tests whether a cognitive measurement can help through a different channel: learning using privileged information (LUPI), where the auxiliary measurement is available only during training. The planned five-arm ZuCo intervention is gated on two preconditions: natural-reading gaze must contain information about the target label, and paragraph-disjoint splits must support a stable low-sample learning curve. The data comprise 300 natural-reading sentences from 12 participants, but only seven source paragraphs [E024].

The gaze measurement itself is reliable: combined split-half reliability is 0.71 [E024]. It is not informative for the target under natural reading. A rich 35-dimensional gaze probe gives binary AUC 0.508 with permutation $p = 0.45$, and five-way relation-type balanced accuracy is 0.202 against 0.200 chance, $p = 0.35$ [E024]. Linear, nonlinear, participant-stacked, and within-

participant probes give the same null pattern. Under task-directed reading, the relation-type probe reaches 0.295 against 0.100 chance, with permutation $p = 0.003$ [E024]. This contrast shows that gaze contains relation information when readers are instructed to search for the relation, but that channel is task-intent leakage rather than portable natural-reading privileged information [E024].

The seven-paragraph structure independently fails the split gate. Across paragraph-disjoint splits, test size ranges from 8 to 161 sentences and the empirical low- n minimum detectable effect ranges from approximately 0.03 to 0.08 or becomes undefined [E024]. No stable leakage-free sample-efficiency estimand is available on this substrate. The binding gate therefore stops the five-arm intervention before training. This is evidence that the chosen privileged signal and dataset fail the required preconditions, not evidence that learning using privileged information is universally ineffective. E009 and E024 fail at different locations and should not be collapsed into a single downstream null. E009 implements downstream evaluation but lacks a reliable brain-specific mediator, so it cannot reveal what a successful alignment change would do. E024 never reaches the intervention because the clean natural-reading privileged signal is neither label-informative nor stably evaluable under paragraph-disjoint splits. Together they show why practical claims require a verified chain from target information to representation movement to held-out task performance. They do not establish that every neural or cognitive signal lacks practical value.

4.6 Separating synthetic-target learnability, control identification, and biological transfer

E020 addressed a separate ceiling question by subtracting a leave-one-participant-out estimate of the shared response. The reference was too unreliable and leakage-sensitive to estimate a general ceiling, so E020 is not part of the identified intervention chain. Appendix B.7 reports the failed instrument and the control that removed its apparent residual signal [E020].

E016 then asks a different question: whether a dense stimulus-derived brain target can be optimized at scale and transfer to measured fMRI alignment. TRIBE v2 is a pretrained multimodal model that predicts high-resolution cortical fMRI responses for new stimuli [28]. E016 uses its 20,484 predicted cortical values per text stimulus as a synthetic neural target, not as newly recorded fMRI. TRIBE and a projected teacher-hidden-state control each define their own synthetic target; KD-only and block-permuted variants provide within-target references. The two synthetic target spaces are dimension-matched in the training harness, but E016 did not establish that they were matched in information, geometry, baseline difficulty, or optimization [E016].

Across six training seeds, TRIBE-target training improves its held-out synthetic target R^2 over KD-only by +0.078298. The projected text-feature target improves its own endpoint by +0.000903 [E016]. The maximum relative perplexity deviation across the checked target and control arms is 0.009220 [E016]. The supported within-target claim is therefore that TRIBE training improves prediction of its own target under nearly unchanged language quality. The larger TRIBE gain than the sentence-local projected teacher-hidden-state gain is a numerical cross-target contrast whose content-specific interpretation requires a comparable control.

The saved students are then scored on one common Tuckute endpoint. Synthetic training uses layer 6, while the real-brain evaluation uses layer 7 and the same five-participant/five-ROI average for every seed [E016]. On the common endpoint, the direct TRIBE-minus-KD mean is -0.000919 , with five of six seed margins negative [E016]. The predeclared relative diagnostic, $[\text{TRIBE} - \text{KD}] - [\text{textfeat} - \text{KD}]$, is -0.001213 ; all six seed margins are negative, giving exact sign $p = 0.03125$ [E016]. Because E026 later shows that the text-feature arm fails measured-axis comparability, this relative diagnostic cannot identify target content. The corresponding difference relative to the two block-permuted gains is -0.001005 , with five of six seed margins negative [E016]. The block permutation leaves 0–30% of rows fixed and duplicates or omits

one row in five of six seeds, so that contrast is a sensitivity analysis rather than an exact-null comparison.

The direct E016 result is a fixed-endpoint transfer failure. The synthetic target is learnable, but its improvement does not improve alignment on this fixed real-brain diagnostic. The inference unit is six stochastic training seeds over one endpoint that already averages five participants and five ROIs. It supports neither participant-level nor population-level inference, and it does not show that all synthetic brain targets must fail. This is stronger than a failure to fit the proxy because target fit succeeds in every seed. It remains narrower than a biological ceiling because the endpoint is fixed, participant-averaged, and evaluated at a different student layer.

E026 tests the missing control-identification condition without retraining and without loading or consuming participant-outcome data [E026]. Under the frozen 55-check conjunction, 30/55 checks fail; 24 pass and one is unresolved [E026]. The mismatch begins before intervention: mean KD-only target R^2 is 0.595327 versus 0.911687 for TRIBE and textfeat, respectively, leaving 4.582 times as much mean headroom for TRIBE. Across two disjoint target samples, normalized covariance-spectrum log-RMS differences are 5.524 and 5.535, and absolute stable-rank log-ratios are 0.634 and 0.616; all exceed their predeclared margins. Low-level nuisance predictability differs by 0.192 and 0.183 held-out R^2 across the two coordinate samples. After training, all six global relative-parameter-update ratios and all six layer-6 centered relative-Frobenius movement ratios fail their margins. This is not total non-parity: standardized global scale, most valid within-document autocorrelation checks, power-spectrum shape, and static-unique target predictability pass their margins, while four of six CKA-distance checks pass. Initial-gradient comparability is unknown because the required values were not retained, and lag 1024 is unresolved because no valid within-document pairs exist.

The audit therefore shows that the saved text-feature arm fails the predeclared measured-axis comparability conjunction; it does not establish universal or information-theoretic incomparability. Because textfeat is a Gaussian projection of the same GPT-2-medium teacher representation used for KD, its high KD baseline and low remaining headroom are consistent with same-teacher redundancy; this is a mechanism hypothesis rather than a causal result. E026 does not identify which mismatch produced the cross-target gain, show that TRIBE contains unique biological information, or explain E025’s transfer failure. It removes the raw TRIBE-minus-textfeat proxy-gain difference as evidence of brain specificity while leaving E016’s within-target learnability result and E025’s near-zero direct TRIBE-minus-KD contrast intact.

E025 then removes the participant-averaging limitation by rescoring the same saved students separately for the nine complete Tuckute participants [E025]. Before opening the participant result, it verifies language quality, target uptake, and retained-student movement. All three gates pass: every target-learning seed is positive, every target-trained student moves detectably from its KD initialization, and the largest direct target/KD or between-target quality difference is 0.002579 bits per byte [E025]. The endpoint is therefore interpretable as a transfer result rather than a failed manipulation.

At fixed layer 7, the predeclared participant-first TRIBE-minus-text-feature mean is +0.000136, with 95% CI $[-0.000505, +0.000777]$ and a one-sided 95% upper bound of +0.000653 [E025]. That upper bound is below the internal +0.002 continuation threshold, and the observed 80% minimum detectable effect is 0.000890 [E025]. The threshold governed whether to continue a positive-mechanism program; it is not a universal minimum biological effect. Only four of nine participant estimates are positive, the median is -0.000023 , and removing the highest-KD-alignment participant changes the mean to -0.000131 [E025].

After E026, the relative contrast remains a valid numerical comparison but cannot identify target content. The direct TRIBE-minus-KD contrast is the cleaner biological transfer endpoint, but it estimates the net effect of the saved TRIBE-training arm versus KD, not brain-derived content in isolation. Its mean is -0.000014 , with 95% CI $[-0.000739, +0.000711]$; seven of

nine participant estimates are negative [E025]. The small positive relative mean partly reflects a negative text-feature-minus-KD mean of -0.000150 rather than a positive direct TRIBE gain [E025]. Imageability adjustment remains near zero, while the broader nuisance stress test and the layer-6 sensitivity are negative [E025].

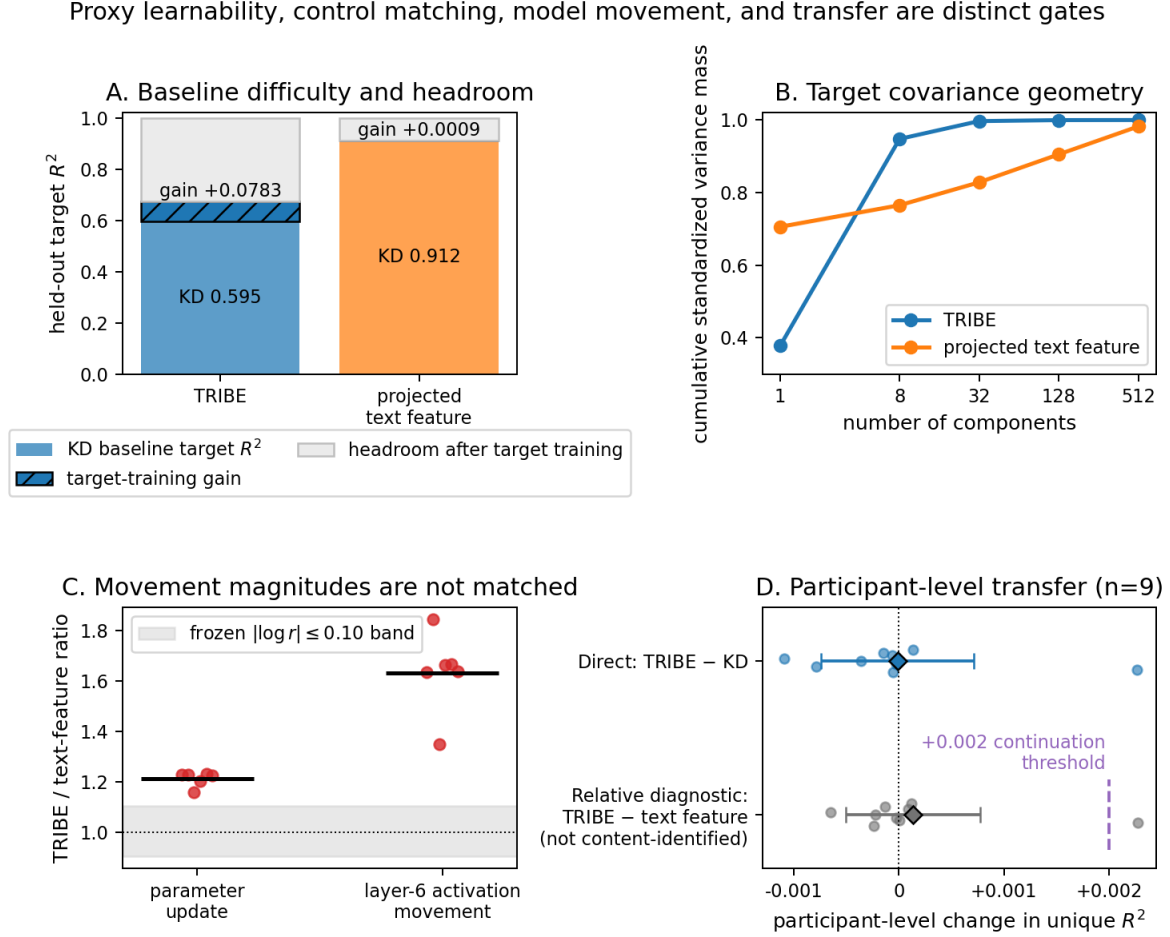


Figure 8: The dense-proxy intervention separates four evidential gates. Panel A distinguishes KD baseline target fit, target-training gain, and remaining headroom. Panel B plots mean cumulative variance mass across the two fixed deterministic samples; shading spans their two values and is not a confidence interval. Panel C shows six-seed global-parameter and layer-6 centered-Frobenius movement ratios against the frozen $|\log r| \leq 0.10$ band; centered-CKA comparability is mixed and reported in the text. In Panel D, dots are participants, diamonds are participant means, and horizontal bars are participant-level mean 95% t confidence intervals. The purple segment marks the internal $+0.002$ continuation threshold for the relative contrast only. The relative estimate cannot identify target content after E026, while the direct estimate remains interpretable as the net TRIBE-training-versus-KD endpoint rather than a content-specific effect [E016][E025][E026].

The four gates support a more specific conclusion than either “the proxy worked” or “brain guidance failed.” Proxy fit and retained-student movement pass. The saved cross-target comparator fails identification. The predeclared relative estimate has a one-sided 95% upper bound below the internal continuation threshold, while the direct participant-level TRIBE-minus-KD estimate is near zero [E025][E026]. This dissociation is restricted to the saved students, fixed layer and nuisance design, nine-participant Tuckute cohort, and recorded sentence/ROI endpoint.

5 Discussion

5.1 Main empirical answer

The study gives different answers to measurement and intervention. Under the recorded controls, trained middle-layer representations predict held-out brain responses beyond the nuisance set and same-architecture untrained networks [E002][E006]. Plain distillation leaves a student below its teacher on the alignment measure, but that gap remains confounded with language-model quality [E003][E015]. These findings establish a usable readout and measurable headroom, not a training benefit.

The dense-proxy branch separates four stages in one saved-student design: target uptake, student movement, comparator validity, and transfer to recorded fMRI. TRIBE-target prediction improves over KD by +0.078298, the retained students move, and language quality remains within the declared tolerance [E016][E025]. The saved text-feature comparator then fails 30/55 frozen checks, so the cross-target proxy advantage cannot identify neural target content [E026]. On recorded fMRI, direct participant-level TRIBE-minus-KD is -0.000014 , with 95% CI $[-0.000739, +0.000711]$. The preregistered relative estimate has a one-sided 95% upper bound of +0.000653, below the internal +0.002 continuation threshold, but that relative estimate is not content-identifying after E026 [E025][E026].

The recorded-target experiments reach the same practical boundary by other routes. The tested MSE, contrastive, higher-capacity, voxelwise, and full-fine-tuning variants do not establish a participant-level benefit, and the downstream experiments do not establish a practical payoff [E004][E008][E009][E011][E013][E017]. In these models and datasets, linear brain alignment is measurable, but the tested alignment objectives do not provide a reliable training signal for the studied compression regimes. This does not imply that neural supervision is ineffective in every model, modality, dataset, or optimization setting.

The experiments also show why target fit alone is insufficient. An auxiliary target can become easier to predict because it offers more headroom or a different covariance geometry. Repeated success across seeds measures optimization repeatability, not biological generalization. Comparable language quality removes one alternative explanation but does not identify the content of the changed representation. A training claim needs an appropriate target control, inference at the biological unit named by the claim, and transfer to a recorded endpoint that was not optimized during training.

5.2 Where the evidence chain breaks

The first break is between measurement and intervention. A linear readout can predict recorded responses from a frozen representation even when the tested auxiliary objectives fail to produce a stable held-out gain. The results are consistent with competition from the language objective, low-sample noisy targets, and language-quality damage under stronger updates, but they do not identify which mechanism caused the failure. The fold-level E004 result, the participant-level E008 result, and the neutral-to-harmful voxelwise trajectory in E013 each locate a failure at this intervention stage under a different readout or inference unit [E004] [E008] [E013].

Increasing flexibility does not by itself resolve the problem. Higher-rank LoRA changes capacity without producing a stronger effective manipulation, a contrastive ROI objective does not rescue the participant-level effect, and full fine-tuning ranges from substantial language-quality damage to a near-zero real-minus-control contrast [E011][E013][E017]. These results implicate the tested induction mechanisms rather than one adapter rank or one loss parameterization. They do not rule out objectives with different targets, architectures, temporal models, or optimization regimes.

The second break is between optimizing a proxy and transferring to biology. E016 shows that a dense brain-derived synthetic target can become substantially easier for a student to predict

under nearly unchanged language quality, while the same students do not improve on a common real-brain endpoint [E016]. E026 locates an earlier identification break: the saved projected text-feature control differs materially in target spectrum, effective rank, KD predictability, remaining headroom, nuisance predictability, and induced movement, so the between-target proxy-gain difference cannot isolate brain-derived content [E026]. E025 resolves the endpoint at the participant level: all manipulation gates pass, yet direct TRIBE-minus-KD is near zero and the preregistered relative upper bound is below its internal continuation threshold [E025]. The direct contrast estimates the net TRIBE-training effect against KD, not brain-derived content in isolation. Together, these results show that the proxy was learnable and changed the student, but do not show target-content specificity or biological transfer.

5.3 Inference unit and response averaging

The inference unit changes the scientific answer because different repetitions support different claims. Pooling three seeds over five shared stimulus folds produced an apparently precise E004 interval, but the valid fold-level analysis has five units and includes zero [E004]. Likewise, the participant-averaged target in E005 supports an apparent gain, whereas the corresponding effects across nine individually evaluated participants center near zero in E008 [E005] [E008]. Seeds measure optimization variability, folds measure stimulus variability, and participants measure biological variability; one axis cannot substitute for another. E025 follows that rule by averaging seeds and folds within each person before forming its nine-participant interval [E025].

Response averaging does not estimate the same quantity as an average participant effect. It suppresses idiosyncratic noise and emphasizes the component evoked consistently by the shared stimulus, thereby increasing reliability and changing the estimand. The nested E010 curve suggested a dose response, but that pattern did not survive random-subset reanalysis, so it cannot carry a causal averaging law [E010]. E014 independently shows that averaging can legitimately increase an encoding score when individual signals are already positive, through higher signal-to-noise ratio and a shift toward the shared component [E014]. The positive averaged-target intervention result therefore does not establish a mean per-person effect in this cohort. At the same time, the higher score on an averaged encoding target is a valid measurement of a different, more shared response.

5.4 What the negative results establish

The negative findings constrain particular mechanisms at particular inference units. E013 establishes that the tested voxelwise MSE distillation objective does not improve held-out alignment across its weight range; it does not establish that naturalistic voxelwise objectives are impossible [E013]. E009 detects no out-of-domain perplexity benefit in a regime where the prerequisite representation contrast is unreliable, so it does not estimate the payoff of a successful alignment manipulation [E009]. E024 stops before a five-arm privileged-information intervention because natural-reading gaze is uninformative about the task label and the paragraph structure cannot support the intended learning curve; it is a failed precondition, not a completed LUP null [E024]. E020 finds that an apparent participant-residual signal is explained by stimulus leakage and autocorrelation under its strongest diagnostic, but its noisy reference prevents the experiment from closing a universal stimulus-predictability ceiling [E020].

The reduced external and cognitive-target experiments have similarly narrow scopes. The faithful-as-feasible Negi-style head in E019 produces no positive brain-specific gain: gentle tuning barely moves the model, whereas stronger tuning damages both perplexity and encoding [E019]. Differences in architecture, dataset, training duration, and metric prevent that result from adjudicating the original Negi study; it instead corroborates the mechanism failure observed in the primary experiments. In E021, a phase-randomized target that preserves the reading-time signal’s autocorrelation and marginal shape reproduces its auxiliary-regularization

advantage, so the gain is attributable to signal shape rather than cognitive content in that setup [E021]. In E022, a reduced single-participant speech experiment does not reproduce the predeclared phoneme benefit, making the planned matched-quality follow-up vacuous, but the reduced dataset and protocol do not refute the multi-participant Moussa result [E022].

5.5 Requirements for future brain-guided training

Future intervention studies can screen for identification failures before expensive training. Before training, the alignment measure should be tested for leakage, against a capacity-fair nuisance model, and against an untrained network. If response units are selected for reliability, the selection must not use the test data. Checks for a proposed target and control should reflect how they were constructed. Relevant examples from E026 include covariance spectrum, effective rank, baseline predictability, remaining headroom, nuisance predictability, initial loss, and gradient scale. Equal output width is not enough.

Training success and outcome success should be reported separately. Target uptake, parameter change, representation movement, and language quality establish that the intervention occurred. They do not establish that the change reflects neural content. That claim requires an adequately matched control and a common recorded-brain endpoint. If the claim concerns people, seeds and folds must be aggregated within each participant before population inference.

A synthetic or averaged target also needs transfer beyond the target used to train it. A direct target-trained-versus-KD contrast can test the net intervention effect even when a cross-target comparator fails, but it cannot by itself isolate which target property caused the effect. Downstream utility should be tested only after the biological mediator is established. This order keeps a failed manipulation, an invalid comparator, and a failed transfer endpoint from being reported as the same kind of null.

5.6 Relation to prior positive studies

Positive intervention studies answer related but different questions. Speech studies report gains after adapting retained encoders to fMRI or ECoG targets, and Vaidya et al. test transfer from fMRI tuning to a newly fitted ECoG readout in different patients [8, 10, 17]. In text, Merlin et al. compare brain-prediction tuning with tuning on the same stimulus text, while Xiao et al. use task-fMRI-derived directions to change reasoning behavior [9, 15]. These results show that neural objectives can be useful under their own designs.

The present experiments differ in model budget, target, comparator burden, and inference unit. They train a smaller student, require language-quality control, audit whether the nonbrain target supports content attribution, and evaluate participant-level transfer. Conversely, the local negative results do not reproduce the datasets, participant pools, or complete optimization procedures of the positive studies. There is no direct contradiction: each result applies to its own intervention and endpoint.

The useful comparison is therefore between estimands rather than headline scores. Measurement, intervention, participant generalization, and downstream payoff require different controls. A result at one stage remains informative, but it does not inherit the claim licensed by a later stage.

6 Limitations

6.1 Biological generalization

The voxelwise measurability result is conditional on one deeply sampled participant, UTS03. Spatially dependent voxels increase resolution but do not provide population replication, and the five story folds share training stories [E006]. The primary intervention generalization tests

include nine participants exposed to the same 1,000 sentences, so participant diversity and stimulus diversity are both limited [E008][E025]. The three-participant LeBel intervention variants are existence probes rather than population estimates [E013] [E017] [E019]. These limitations are asymmetric: a controlled within-participant positive can establish that a signal exists under that participant’s measurement process, while a within-participant null cannot establish absence across the population. A broader population estimate would require more independently sampled people, prespecified participant-level aggregation, and enough precision to resolve between-person heterogeneity.

6.2 Stimulus and target coverage

The Tuckute benchmark contains isolated English sentences summarized into five left-hemisphere language ROIs, whereas the LeBel experiments use naturalistic English stories from a small number of deeply sampled listeners. Neither substrate represents the full range of linguistic contexts, modalities, languages, or cortical systems. The isolated-sentence benchmark favors sentence-local representations and removes discourse history, whereas story listening introduces temporal context, acoustic structure, and hemodynamic alignment. Agreement across them is useful robustness evidence, but disagreement would not by itself identify whether stimulus format, modality, target resolution, or participant sampling caused the difference. The gaze experiment uses only 300 natural-reading sentences from seven highly unequal paragraphs, which prevents a stable paragraph-disjoint sample-efficiency curve [E024].

6.3 Construct validity

Unique R^2 is conditional on the nuisance variables actually included. The pipeline subtracts rate, duration, frequency, and static lexical or semantic features, but it deliberately leaves surprisal and imageability outside the primary nuisance set; “beyond nuisance” therefore does not mean beyond all stimulus-predictable content [E006][E025]. The trained-minus-untrained gap identifies what language training adds over a random network under those controls, not shared cognitive mechanism or representational identity. E020 further shows that estimating a stimulus-predictability ceiling is limited by the reliability of the empirical reference [E020].

6.4 Intervention strength

The tested objectives cover MSE, a contrastive ROI loss, LoRA capacity changes, full fine-tuning, behavioral auxiliary targets, and one dense synthetic target, but they do not span all possible neural objectives or optimization schedules. In E011, additional adapter capacity does not produce a materially stronger representation change, while E013 and E017 show that more aggressive intervention can instead damage the model [E011] [E013] [E017]. A failed manipulation cannot identify the consequence of a successful one. Conversely, a manipulation that strongly fits its training target can still fail to improve an independent biological endpoint, as the synthetic-to-real transfer gate demonstrates. Future objectives need both a manipulation check on the optimized target and an independent biological endpoint that was not used to construct that target.

6.5 Inference and power

E004 has five valid seed-averaged fold units, and its corrected interval cannot exclude a small effect near the observed mean [E004]. Several sensitivity analyses reuse folds, stories, or participants and are therefore not independent replications. The block-permutation implementation can leave some target rows fixed, so it is treated as a sensitivity control rather than an exact marginal-preserving randomization null [E016]. Minimum-detectable-effect calculations bound

the tested contrasts but do not convert a null into evidence of exact zero. E008’s participant-level 95% interval excludes a $+0.003$ unique R^2 effect, and its post-run calculation reports power 1.0 at that reference effect size [E008]. For E025, the preregistered relative one-sided upper bound is below the internal $+0.002$ continuation threshold. The direct TRIBE-minus-KD 95% interval also lies entirely below $+0.002$, although that threshold did not define the direct contrast [E025]. That threshold decided whether to continue the positive-mechanism program; it is not a universal minimum biological effect. Smaller effects and effects in other regimes remain possible. Other nulls are correspondingly narrower when their valid unit count is smaller or when the intervention itself fails to move the representation.

6.6 External reproduction

E019 changes the architecture, dataset, training duration, and some implementation details relative to Negi, so it provides corroboration on the available substrate rather than a faithful refutation [E019]. E022 uses one participant and a reduced speech setup rather than the full multi-participant Moussa protocol, so its non-reproduction is likewise local [E022]. These experiments test whether the reported mechanisms transfer into the available controlled pipeline; they do not estimate the original studies’ effects.

6.7 Synthetic-target comparability

TRIBE and textfeat have the same nominal output width, but E026 finds that the saved targets fail predeclared comparability criteria for covariance geometry, effective rank, KD baseline predictability and headroom, low-level nuisance predictability, global parameter displacement, and centered-Frobenius representation movement [E026]. Their held-out target R^2 values are therefore meaningful within target family, not as a content-identifying shared cross-family scale. Some measured properties do match, including standardized global scale, most tested order and power summaries, and static-unique predictability; initial-gradient matching remains unknown, and the audit never tests information equivalence. E016’s original transfer endpoint averages five participants and five ROIs, and its synthetic objective and transfer readout use different student layers. E025 rescoring supplies participant-level inference for nine complete participants, but all participants share one isolated-sentence corpus and the same five language ROIs [E025]. The measured-axis failure settles that the saved text-feature arm cannot identify brain-specific content in the cross-target gain comparison, while the defective permutation remains unsuitable as an exact target-specific null. The direct TRIBE-minus-KD contrast remains interpretable as the net effect of the saved TRIBE-training arm on the recorded endpoint, but it does not isolate brain-derived content from other properties of that target.

6.8 Downstream-task coverage

The primary practical test measures out-of-domain perplexity rather than task accuracy, robustness, calibration, or sample efficiency across a broad benchmark suite [E009]. Because its alignment manipulation is unreliable, the test does not identify the downstream value of a successful brain-specific representation change. The gaze-privileged route stops before intervention and the reduced speech route fails its reproduction gate, leaving downstream utility bounded by preconditions rather than exhaustively tested [E022] [E024].

7 Conclusion

This thesis asked whether nuisance-controlled linear brain alignment can do more than describe a trained language model: can it guide the training of a smaller student? The measurement premise holds in the tested settings. Trained middle layers predict held-out fMRI beyond the

recorded nuisance variables and untrained-network controls [E002][E006]. Plain distillation also leaves measurable alignment headroom, although the amount attributable to distillation rather than language-model quality remains unresolved [E003][E015].

The intervention premise was not supported in the tested regimes. Recorded-target objectives do not establish a reliable participant-level gain, and changing adapter capacity, loss form, target resolution, or trainable parameter count does not rescue the result [E008][E011][E013][E017]. The dense-proxy experiment passes target-uptake, student-movement, and language-quality gates, but its saved cross-target comparator fails measured-axis identification [E016][E025][E026]. On recorded fMRI, direct participant-level TRIBE-minus-KD is -0.000014 , with 95% CI $[-0.000739, +0.000711]$. The preregistered relative comparison has a one-sided 95% upper bound of $+0.000653$, below the internal $+0.002$ continuation threshold, but E026 prevents that comparison from identifying neural target content [E025][E026].

These results are limited to the models, objectives, participants, stimuli, and endpoints studied here. They do not show that neural supervision is universally ineffective. They show that a positive encoding score, successful proxy fit, or repeated optimization effect is not enough to establish a useful brain-guided intervention.

Future claims should separate measurement, target uptake, comparator assessment, biological transfer, and downstream utility, and define the estimand for each quantitative stage. Language quality must be controlled, technical repetitions must not replace biological units, and synthetic or averaged targets must transfer to independently recorded responses. Under those requirements, brain alignment remains a measurable readout under the stated controls in this study, but not a demonstrated reliable training signal for the tested compression regimes.

A Experiment coverage and disposition

This appendix accounts for every experiment identifier used by the project, including engineering pilots, unrun designs, failed precondition gates, and intentionally unused identifiers. An experiment is classified as scientific evidence only when it produced an interpretable result for a stated estimand; implementation activity alone is not treated as evidence. Table 6 gives the complete disposition, while Appendix B expands the secondary experiments and failed instruments that would interrupt the main causal sequence.

Table 6: Complete experiment coverage and disposition.

ID	Question	Execution state and verdict	Location and boundary
E001	Recover a planted synthetic relationship?	Implemented; engineering validation only.	Appendix B.1; no real-neural claim.
E002	Predict Tuckute responses beyond nuisance and untrained features?	Complete; controlled ROI measurability pass.	Section 4.1; raw unique R^2 , no invalid ceiling fraction.
E003	Retain alignment under distillation?	Complete; quality-entangled retention gradient.	Section 4.2; objective-specific shedding unidentified.
E004	Move held-out alignment with a brain loss?	Complete and reanalysed; no demonstrated gain at five-fold inference.	Section 4.3; 15-cell bootstrap superseded.
E005	Beat a matched-perplexity control on an averaged target?	Complete and reanalysed; small averaged-target trend.	Section 4.3; Appendix C.4; original excludes-zero claim superseded.
E006	Extend measurability to naturalistic voxels?	Complete and corrected; conditional UTS03 pass.	Section 4.1; voxel-population intervals and derived MDE retracted.
E007	Power calculation following E006?	Identifier intentionally unused.	No experiment record; invalid calculation never promoted.

ID	Question	Execution state and verdict	Location and boundary
E008	Generalize the intervention across participants?	Complete; no demonstrated participant-level gain.	Section 4.3; Appendix C.3; nine participant units, with post-run power 1.0 at +0.003.
E009	Improve out-of-domain perplexity?	Complete; bounded practical null with an unreliable mediator.	Section 4.5.
E010	Increase the contrast by averaging more participants?	Complete including random-subset re-run; no identified dose response.	Section 4.3; Appendix B.2; nested interpretation superseded.
E011	Rescue the result with greater adapter capacity?	Complete; higher rank and duration did not strengthen the manipulation.	Section 4.4.
E012	Run a voxelwise participant-level study?	Deferred and unrun because three participants could not support the claim.	Coverage only.
E013	Rescue the intervention with contrastive or voxelwise objectives?	Complete for tested branches; ROI null and voxelwise manipulation failure.	Section 4.4; larger expansion unbuilt.
E014	Improve encoding by averaging without intervention?	Complete; legitimate SNR improvement and estimand shift.	Section 4.3; Appendix B.3; no universal artifact claim.
E015	Relate model quality to alignment?	Complete and corrected; strong full-range, moderate uncertain capable-band coupling.	Section 4.2; Appendix C.5; $r \simeq -0.92$ superseded.
E016	Train on a dense synthetic neural target and transfer?	Complete and independently reviewed; synthetic positive, fixed real-brain transfer negative.	Section 4.6; Appendix B.4; dimension-matched control and seed-unit scope.
E017	Produce a gain with full fine-tuning?	Complete; three-participant existence probe found no specific gain.	Section 4.4; Appendix C.6; no nine-cell population inference.
E018	Reserved identifier?	Intentionally unused.	No scientific content.
E019	Reproduce a local Negi-style gain?	Complete; the local objective did not establish a gain.	Section 5.6; Appendix B.6; not a Negi refutation.
E020	Estimate a stimulus-predictable residual ceiling?	Complete; failed instrument with bounded controlled residual.	Section 4.6; Appendix B.7; full ceiling remains open.
E021	Is reading-time benefit cognition-specific?	Complete through phase-randomized control; shape-specific, not cognition-specific.	Section 5.6; Appendix B.8; earlier reads superseded.
E022	Reproduce a Moussa-style speech gain?	Complete pilot; reduced single-participant non-reproduction.	Section 5.6; Appendix B.9; not a full-study refutation.
E023	Identify objective-specific shedding against a quality frontier?	Design plus analysis-only slope gate; no training result.	Appendix B.11; predeclared saturated comparison not cleared.
E024	Use gaze as train-only privileged information?	Gaze preconditions complete; intervention stopped for informativeness and split validity.	Section 4.5; Appendix B.10; EEG untested.
E025	Does learnable synthetic-target training transfer across individual participants?	Complete and independently reviewed; direct TRIBE-minus-KD is near zero, and the preregistered relative upper bound is below the internal +0.002 continuation threshold.	Section 4.6; nine participant units; E026 prevents target-content attribution from the relative comparison.
E026	Are the saved TRIBE and text-feature targets comparable on geometry, learnability, and movement?	Complete and independently audited; 30/55 frozen measured-axis checks fail and one is unresolved, so the saved comparator is non-comparable on measured axes.	Section 4.6; deterministic geometry and six technical seeds do not establish information equivalence or a causal mechanism.
E027	Train one matched nonbrain target?	Not authorized because E025 did not pass the internal continuation rule.	No experiment; positive-mechanism branch closed for this program.
E028	Falsify a reported cross-person fMRI-to-ECOG intervention externally?	DESIGN HOLD after a metadata-only inventory; no endpoint data opened and no training authorized.	Conference extension only; author artifacts, executable boundaries, dimension benchmark, anti-confound review, and final oracle remain gates.

ID	Question	Execution state and verdict	Location and boundary
E029	Can an endpoint-directional certificate prospectively distinguish beneficial, null, and harmful auxiliary targets?	DESIGN HOLD; algebra validator only, no scientific result.	Coverage only; generic theory is occupied and activation depends on a prospective controlled panel plus a surviving E028 positive.

A.1 Supersession map

Five corrections materially change how earlier results are described. First, E004 and E005 originally treated seed-fold cells as independent observations; current inference is performed at the fold unit, and E008 supplies the participant-level test. Second, E006 originally bootstrapped thousands of spatially dependent voxels as if they were independent biological replicates; the retained result is a conditional one-participant point contrast with five held-out-block sensitivity checks. Third, E010’s nested participant averages suggested a monotone dose response, but the E010b random-subset analysis showed that subset identity and target construction could explain the pattern. Fourth, E015’s initial eight-model correlation of approximately -0.92 was replaced by a tokenizer-independent 22-model, six-family analysis with a weaker capable-band relationship and an effective-generation limitation. Fifth, E021’s apparent reading-time advantage survived ordinary learnability controls but disappeared against a content-free control matched for autocorrelation and marginal shape; the final claim concerns signal shape rather than cognition.

The Tuckute noise-ceiling correction is a scale correction rather than a changed experimental result. The stored value 0.491 is a correlation ceiling, whereas unique R^2 is on a variance scale, so earlier ratios of unique R^2 to 0.491 are invalid and are not used here. [E002] The raw unique- R^2 values, trained-minus-untrained contrasts, intervention contrasts, and their substantive verdicts are unchanged.

B Secondary experiments and failed instruments

This appendix explains experiments that validate the apparatus, diagnose a main-track result, test a failed measurement instrument, or provide a reduced external reproduction. Each subsection states the question, intervention or comparison, estimand and inference unit, result, conclusion, limitation, and role in the thesis.

B.1 E001: end-to-end implementation validation on a planted synthetic signal

E001 asked whether the initial implementation could recover a known relationship from synthetic language features and synthetic fMRI-like targets. The run exercised feature extraction, ridge fitting, cross-validation, and result serialization while holding the planted data-generating relationship under experimenter control. Its estimand was recovery of that constructed relationship, and the relevant unit was the synthetic sample rather than a participant or brain measurement. The implementation recovered the planted signal and therefore passed its engineering validation. [E001] This result licenses use of the implementation as a starting point, but it provides no evidence that a language model predicts real neural activity, that the nuisance controls are sufficient on real stimuli, or that a neural objective can improve training. E002 is the first scientific evidence because it replaces the constructed target with recorded language-network responses and adds trained-versus-untrained and nuisance controls.

B.2 E010 and E010b: participant averaging as a target intervention

E010 asked whether the apparent brain-specific intervention contrast increased as more participants were averaged into the target. The motivation was the discrepancy between E005, which used a five-participant average, and E008, which found a near-zero mean across individual participants. For nested averages of $k \in \{1, 2, 3, 5, 9\}$ participants, the measured intervention gaps were approximately $-0.0002, -0.0001, +0.0023, +0.0194, +0.0071$. [E010] The original interpretation treated the rising $k = 1$ -to-5 limb as a reliability dose response while noting the reversal at $k = 9$. That interpretation was not identified because every larger target contained the smaller target, participant order was fixed, and target construction changed together with signal-to-noise ratio.

E010b therefore repeated the analysis over random participant subsets at each k . The resulting subset distributions overlapped substantially and did not support a stable monotone causal relation between participant count and the intervention contrast. [E010] The estimand in both analyses was the contrast for a constructed group-average endpoint, while participant subsets, not seed-fold cells, supplied the sensitivity variation. The joint conclusion is that averaging changes both reliability and the biological target, and the nested curve does not establish that increasing k causes a brain-specific training effect. Its role is diagnostic: it explains why E005 and E008 can differ without treating either endpoint as universally correct.

B.3 E014: averaging improves encoding signal without establishing an intervention artifact

E014 separated the effect of participant averaging on the encoding measurement from its effect on brain-guided training. It held the model features and encoding procedure fixed while replacing individual targets with averages over increasing numbers of participants. Raw encoding unique R^2 rose across $k = 1, 2, 3, 5, 9$ as $+0.0039, +0.0162, +0.0324, +0.0444, +0.0702$. [E014] This is the expected consequence of suppressing participant-specific and measurement noise in a shared stimulus-evoked target. The estimand is nevertheless different at each k : an individual response at $k = 1$ and an increasingly group-shared response as k grows. E014 therefore supports participant averaging as a legitimate way to improve the signal-to-noise ratio of a group-level encoding target. It does not show that averaging universally manufactures a false encoding score, and it does not rescue participant-level inference for E005. Its role is to keep the SNR benefit separate from the biological-generalization question answered by E008.

B.4 E016 prerequisite gates and the synthetic-to-real transfer test

E016 asked whether a dense stimulus-predictable neural proxy could overcome the scarcity of measured fMRI targets and provide a usable auxiliary training signal. The program used TRIBE v2 to generate a high-dimensional synthetic neural target from text-associated stimuli, trained language-model students with that target, and then evaluated whether the resulting improvement transferred to an independent real-brain endpoint.

The first prerequisite was fidelity: a synthetic target is useful only if it captures stable stimulus-dependent structure and can be predicted by the selected student representation. The fidelity analyses showed that the proxy was learnable enough to support a controlled training experiment, but a separate attempt to use it as a formal ceiling on real-brain alignment failed because proxy predictions captured only a limited and target-dependent portion of recorded response variance. [E016] The ceiling analysis is therefore a failed instrument rather than evidence that no additional stimulus-predictable neural structure exists.

The training comparison used six stochastic seeds and the same KD backbone across target arms. On their respective synthetic endpoints, the TRIBE-guided arm improved target R^2 over KD by $+0.078298$, whereas the sentence-local projected teacher-feature control improved by

+0.000903. [E016] E016 established only output-dimension matching for the text-feature target; it did not itself establish information, geometry, baseline-difficulty, or effective-rank matching. The block-permuted control also retained some fixed points and is described as block-permuted rather than fully deranged.

The independent transfer gate scored saved students on the fixed Tuckute target averaged over five participants and five functional ROIs. The inference unit was the six stochastic training seeds for this one endpoint, not participants. Direct TRIBE-minus-KD was -0.000919 , with five of six seed estimates negative [E016]. The relative TRIBE-minus-text-feature gain was -0.001213 ; all six margins were negative and the exact sign test gave $p = 0.03125$ [E016]. E026 later shows why that relative value cannot identify target content. The corresponding comparison against permuted-control gains was also negative, and PCA ranks 25, 50, and 100 did not reverse the route. Synthetic training used layer 6 while the Tuckute diagnostic read layer 7, which further bounds mechanistic interpretation. The conclusion is a transfer failure in this fixed diagnostic, not a population-level null and not a universal upper bound on synthetic neural supervision. E016 has a specific role in the thesis: learnability of a neural-looking target does not establish usefulness for an independently measured biological endpoint.

E016 measures proxy uptake and fixed-endpoint transfer separately

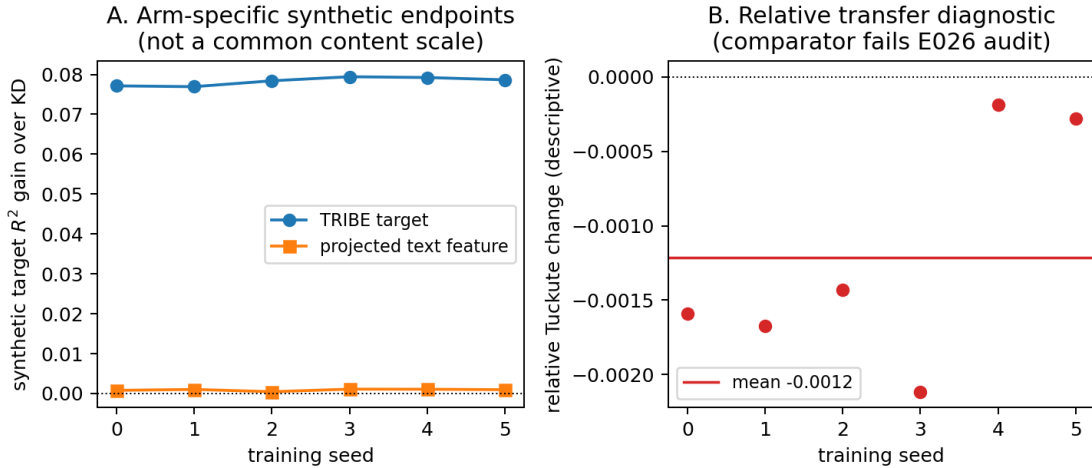


Figure 9: The frozen E016 seed-level diagnostic. Panel A shows target- R^2 gains over KD on each arm’s own synthetic endpoint. Those values are not a common content scale. Panel B shows the relative TRIBE-minus-text-feature change on one fixed participant-averaged Tuckute endpoint. All six values are negative, but the text-feature comparator later fails the E026 measured-axis audit. E025 supplies the separate participant-level result [E016][E025][E026].

B.5 E026: retrospective target-control identification audit

E026 audits the existing E016 targets and students without retraining and without using Tuckute participant outcomes [E026]. Two disjoint deterministic 4,096-item samples provide fixed geometry checks, while the six existing seeds quantify only variability of the frozen training procedure. The mechanical conjunction has 55 required measured-axis checks: 30 fail, 24 pass, and the lag-1024 order check is unresolved because there are no valid within-document pairs. Mean KD-only target R^2 is 0.595327 versus 0.911687 for TRIBE and textfeat, respectively, leaving a TRIBE-to-textfeat remaining-headroom ratio of 4.582. Normalized spectrum log-RMS differences are 5.524 and 5.535 across the two samples; TRIBE versus textfeat stable ranks are descriptively 2.658 versus 1.410; 2.637 versus 1.424, with absolute log-ratios 0.634 and 0.616; and low-level nuisance-predictability gaps are 0.192 and 0.183. The spectrum differences, stable-rank log-ratios, and nuisance gaps each exceed their frozen engineering margins.

All six global parameter-update ratios and all six layer-6 centered-Frobenius movement ratios also fail, whereas four of six centered-CKA distance checks pass. Standardized global scale, most valid temporal-order checks, binned power shape, and static-unique nuisance predictability pass. Initial-gradient comparability is unknown because the gradients were not retained. The result shows that the saved comparator fails the predeclared measured-axis conjunction; it does not prove information inequivalence, assign causality to any mismatch, establish brain-specific TRIBE content, or explain E025’s near-zero direct TRIBE-minus-KD contrast. Consequently, the larger cross-target synthetic gain cannot carry a specificity claim, while the within-TRIBE learning gain remains valid.

B.6 E019: external Negi-style objective under local controls

E019 asked whether a faithful-as-feasible implementation of a published Negi-style brain-tuning head would first reproduce a local gain and then survive matched-quality and target-specificity controls. The implementation combined the available naturalistic LeBel substrate with a differentiable temporal head and tested learning-rate, probe, and positive-control variants before the final comparison. The relevant estimand was held-out alignment change for the real target relative to a permuted target under the same training and evaluation path. The head did not produce a stable target-specific improvement on the local substrate, so the planned matched-quality analysis of a reproduced positive could not be performed. [E019] Across three participants and three training seeds, gentle tuning changed encoding correlation by -0.0006 ± 0.0027 and unique R^2 by -0.0001 ± 0.0010 relative to the base model, where the terms after $\pm$ are descriptive standard deviations across the nine participant–seed cells. [E019] The real-minus-permuted specificity contrast was -0.0004 ± 0.0018 , and the real-minus-intended-quality-control contrast was -0.0010 ± 0.0018 on the same descriptive cell scale. [E019] The intended quality control did not achieve a match: the real-target arm had perplexity about 51.1–51.7, while the control was about 40.1–40.3, so the latter contrast is not a matched-quality estimate. [E019] Participants, not the nine participant–seed cells, are the biological inference units; these values are mechanism checks rather than population confidence intervals. Stronger learning rates produced negative encoding changes, and the strongest probe increased perplexity from approximately 50 to 11,514 while reducing encoding correlation by about 0.055. [E019] This corroborates the mechanism failure seen in the thesis’s own objectives because a distinct objective family also failed to establish movement. It does not reproduce the published dataset, participant pool, full training scale, or exact implementation, and therefore cannot refute Negi et al. Its role is convergent boundary evidence: the local null is not confined to one MSE loss, but external validity remains unresolved.

B.7 E020: an empirical stimulus-predictable ceiling that did not close

E020 asked whether an empirical reference constructed from other participants could approximate $\mathbb{E}[Y \mid S]$ and reveal brain response structure beyond the stimulus-predictable mean. The intended comparison residualized a target participant’s response against a leave-one-participant-out group reference, then measured how much language-model signal remained. An initial residual appeared to preserve substantial alignment, which would have suggested a participant-specific component beyond the shared stimulus response. Diagnostics showed that the reference was itself unreliable (0.33), residual split-half reliability was 0.174, and temporal leakage and autocorrelation could preserve apparent signal after subtraction. [E020] Once the cleanest controlled construction was used, the trained-minus-untrained residual gap was -0.018 . [E020] The inference used six participants on one fixed story, and the reference did not provide a high-reliability estimate of the latent conditional mean. E020 therefore rejects this particular ceiling instrument while leaving the general ceiling question open. Its role is to prevent the Discussion from turning a failed measurement into either a positive participant-specific mechanism or a

universal closure result.

B.8 E021: reading-time content versus statistical shape

E021 asked whether human reading time supplies a cognition-specific auxiliary target after removing the component predictable from language-model surprisal. The early comparisons used raw reading time, surprisal-residualized reading time, permuted targets, random targets, word length, frequency, and other-model surprisal, and evaluated frozen-probe learning curves across training seeds. Raw and surprisal-residualized reading time performed similarly, showing that removal of the model’s own surprisal-predictable component did not remove the apparent benefit. [E021] In the penultimate analysis the residual target beat several learnability-matched controls, but those controls did not match its strong temporal autocorrelation and heavy-tailed marginal distribution. The final control phase-randomized the signal while preserving its autocorrelation and approximate marginal shape but removed behavioral content. That content-free shape twin tied the reading-time residual: the difference in area under the frozen-probe learning curve was +5.3 with a 95% interval $[-7.9, +18.5]$, where lower area means faster or better learning. [E021] The independent units were stochastic training seeds, and the claim is restricted to this learning-curve setup and target construction. The conclusion is that the observed advantage is attributable to signal shape or regularization, not demonstrated cognition-specific information. E021’s role is methodological: auxiliary targets require controls matched for learnability and statistical geometry, not only random or permuted labels.

B.9 E022: reduced Moussa-style speech reproduction

E022 asked whether brain-guided tuning of a pretrained speech representation reproduced a reported gain in phoneme classification before investing in a full matched-quality shrink test. The reduced setup used one LeBel participant, 24 stories, a wav2vec2-base initialization for both arms, and three training seeds. The primary estimand was the brain-tuned minus pretrained phoneme-F1 difference on held-out speech. The observed gain was +0.52 F1 points with a 95% interval $[-0.36, +1.40]$, below the predeclared +2-point continuation gate and including zero. [E022] The planned matched-twin shrink test was consequently vacuous and was not built. Because this setup is smaller and differs from the published multisubject experiment, it is a reduced non-reproduction rather than evidence that the published effect is false. Its role is to show that the thesis did not obtain an external positive on which to demonstrate a matched-quality collapse.

B.10 E024: privileged gaze information stopped at the precondition gate

E024 asked whether gaze or EEG observed only during training could improve the sample efficiency of a text classifier, following the learning-using-privileged-information setting. The planned estimand was the area between held-out learning curves for a text-only student and privileged-information arms at paired training subsets. Before building five training arms, the design predeclared gaze as the first binding gate and required it to be reliable, informative about the label under natural reading, and evaluable with a leakage-free grouped split.

The reliable gaze summaries largely tracked sentence length and did not predict the relation label under natural reading. Across binary and multiclass probes, linear and nonlinear read-outs, participant-averaged and within-participant features, natural-reading performance stayed near chance; the rich multiclass probe gave balanced accuracy 0.202 against chance 0.200 with permutation $p = 0.35$. [E024] In task-directed reading, by contrast, gaze predicted relation type with balanced accuracy 0.295 against chance 0.100 and permutation $p = 0.003$. [E024] This contrast indicates task-intent leakage rather than a portable natural-reading signal.

The dataset also contained 300 sentences drawn from only seven paragraphs, so a paragraph-disjoint test set varied from 8 to 161 sentences across split seeds and did not support a stable low-data learning-curve estimand. These two failures were independent: the signal was uninformative for the clean target, and the grouping structure prevented a stable leakage-free evaluation. The five-arm LUPI intervention was therefore stopped at its binding gaze precondition gate, and EEG was not evaluated. This is not a completed negative intervention or an EEG precondition result; it is evidence that the proposed gaze-first substrate cannot identify the intended transferable privileged-information effect.

B.11 E023: objective-specific shedding as an unexecuted identification program

E023 formalized a counterfactual question left open by E003 and E015: whether a distilled model lies below the alignment expected for a normally trained model of the same language-model quality. It proposed a same-lineage quality frontier, a matched-quality KD-versus-ordinary-fine-tuning comparison, and operation in a regime where the frontier is sufficiently flat to separate objective effects from ordinary quality loss. The only executed component was an analysis-only slope and knee gate using already measured Pythia and Qwen models. The Pythia 1B-to-410M pair met the local flatness screen with slope $+0.015$, alignment change $+0.0007$, and quality change 0.066 bits per byte. [E023] The three checked Qwen pair slopes were $+0.334$, -0.217 , and $+0.076$, so none met the predeclared saturated-regime requirement [E023]. The analysis-only gate marked a lower-quality pilot as feasible but did not clear the planned training comparison; no inferential unit or hypothesis test was claimed for this descriptive screen. No KD-versus-fine-tuning intervention was run, so E023 contributes an identification framework rather than a scientific result about objective-specific shedding.

C Formal and statistical detail

C.1 Notation and the assumption-bound relationship between conditional information and unique variance

Let Y denote a scalar brain-response target, X the language-model feature block, and Z the nuisance feature block. Conditional mutual information asks how much statistical dependence remains between X and Y after Z is known:

$$I(X; Y | Z) = H(Y | Z) - H(Y | X, Z). \quad (12)$$

This definition is distributional and does not specify a linear estimator. Under a population multivariate Gaussian model with correctly specified unregularized conditional means, the reduction in conditional residual variance can be written as

$$I(X; Y | Z) = -\frac{1}{2} \log(1 - R_{\text{partial}}^2). \quad (13)$$

Here R_{partial}^2 measures the fraction of the variance left after Z that is explained by adding X .

The empirical score used in this thesis is instead semipartial cross-validated unique variance,

$$\Delta R^2 = R_{\text{full}}^2 - R_{\text{nuisance}}^2. \quad (14)$$

Under the same population assumptions, partial and semipartial coefficients are related by

$$\Delta R^2 = (1 - R_{\text{nuisance}}^2) R_{\text{partial}}^2. \quad (15)$$

The thesis fits regularized ridge models on finite folds and evaluates them out of sample, so neither equality makes the measured ΔR^2 an estimator or lower bound for conditional mutual

information. The information-theoretic formulation supplies a conceptual target, while the actual estimand is incremental held-out linear prediction beyond the specified nuisance design.

The nuisance set is also necessarily incomplete. Length, position, static embeddings, acoustic tiers, and temporal controls block named alternative pathways, but they do not condition on every aspect of stimulus content. A positive ΔR^2 therefore licenses the statement that contextual features add held-out linear predictive value beyond those controls. It does not isolate a neural computation, establish causal equivalence, or distinguish prediction of a shared high-level stimulus component from prediction of participant-specific processing.

C.2 Regularization, data processing, generalization, and the rate–distortion analogy

An auxiliary brain term gives an optimization objective of the form

$$\hat{\theta} = \arg \min_{\theta} \{ \mathcal{L}_{\text{text}}(\theta) + \lambda \mathcal{L}_{\text{brain}}(\theta) \}. \quad (16)$$

The brain term biases optimization among solutions that also fit the text objective. It can be read as a maximum-a-posteriori prior only if $\exp[-\lambda \mathcal{L}_{\text{brain}}(\theta)]$ is normalizable and independent of the observed text likelihood. Without those conditions it is more precise to call it an auxiliary loss or regularizer.

The data-processing inequality states that $I(A; C) \leq I(A; B)$ for a justified Markov chain $A \rightarrow B \rightarrow C$. If a compressed representation C is literally a fixed post-processing of teacher representation B , then its information about a brain variable A cannot exceed the teacher’s. Ordinary knowledge distillation does not automatically instantiate this chain because both teacher and student receive direct stimulus or corpus information. The inequality therefore motivates a preservation question but does not predict the sign of the empirical encoding-score change under training.

For a learning algorithm mapping a sample Z^n to parameters W , a sub-Gaussian information-theoretic generalization bound has the form

$$\left| \mathbb{E}[L(W) - \hat{L}(W)] \right| \leq \sqrt{\frac{2\sigma^2 I(W; Z^n)}{n}}. \quad (17)$$

The right-hand side grows with algorithm–sample mutual information. The bound does not imply that adding a neural signal creates a benefit, that the benefit scales as an added information budget, or that a low-data regime must improve. An auxiliary constraint could tighten this bound only if it reduces dependence on the sampled text while preserving empirical fit, which was not measured here. Earlier order-of-magnitude information-budget calculations are therefore not part of the thesis argument.

Classical rate–distortion theory defines

$$R(D) = \min_{P_{\hat{S}|S}: \mathbb{E}[d(S, \hat{S})] \leq D} I(S; \hat{S}). \quad (18)$$

The empirical experiments do not identify this object because parameter count or FLOPs are not the mutual-information rate and alignment loss is not established as the source-coding distortion. The phrase “rate–distortion-style trade-off” is used only for the engineering Pareto relation among model budget, language-model quality, and measured alignment.

C.3 Inference units and power

An inference unit is the independently varying entity over which an uncertainty statement generalizes. Training seeds quantify optimizer variability, contiguous test folds quantify sensitivity

to stimulus blocks, voxels quantify spatial measurement breadth, and participants quantify biological population variation. These units answer different questions and cannot be substituted by multiplying their counts.

E026 uses two disjoint deterministic item samples to test the stability of fixed target-geometry summaries; they are not population replicates and receive no confidence intervals or p values [E026]. Its six paired seeds are technical realizations of the frozen training procedure. Reported seed intervals are descriptive t intervals, and the exact 2^6 magnitude-preserving sign-flip test relies on exchangeability of paired seed-contrast signs under the sharp null; neither device licenses biological or target-population inference.

E004 and E005 used three seeds and five rotating folds, producing 15 observed seed-fold cells. Because seeds within one fold share the same held-out stimulus block and fold effects dominate the contrast, treating all 15 cells as independent understates uncertainty. E004’s mean Qwen contrast was +0.00316, but the five-fold t interval was $[-0.00227, +0.00858]$, and leaving out the influential fourth fold reduced the mean to +0.0014. [E004] The correct conclusion is no demonstrated fold-level intervention effect, with limited power to exclude a small effect.

E008 instead averaged repeated training variability within each of nine participants and treated participants as the biological inference unit. Its mean was +0.00010 with participant-axis t interval $[-0.00037, +0.00058]$. [E008] A +0.003 reference effect lay well outside that interval, and the post-run calculation reported power 1.0 at that effect size while leaving smaller effects possible. [E008]

E006 illustrates a distinct error. Bootstrapping 11,442 spatially dependent voxels generated a narrow interval but did not create participant replication. The retained evidence is the conditional UTS03 point gap, +0.0207 for GPT-2 and +0.0277 for Qwen, plus positive signs on all five held-out story blocks. [E006] Because those blocks share training stories, even the five signs are sensitivity evidence rather than an independent population confidence interval.

Power calculations must use the variance of the future paired contrast at its intended inference unit. The retracted E007 calculation used unmatched absolute fold summaries from E006 and therefore did not estimate the variance of a future tuned-minus-control effect. No replacement participant count is inferred from it.

C.4 E005 reanalysis and the lambda sweep

E005 compared real-target brain-guided KD with a block-permuted-target twin while matching the text-training setup and approximately matching perplexity. The initial averaged-target estimate was +0.0081. [E005] Its original bootstrap sampled 15 seed-fold cells independently and excluded zero, but the corrected fold-unit analysis showed that one held-out block carried a disproportionate share of the result and the valid interval included zero. The retained description is therefore a small averaged-target trend rather than a confirmed brain-specific improvement.

A later Qwen lambda sweep asked whether stronger auxiliary weighting produced a stable alignment gain without unacceptable language-model degradation. Larger weights moved the training endpoint more strongly but also worsened perplexity, and no setting established a free target-specific improvement at matched quality. [E005] This supports a trade-off description rather than a favorable frontier shift. The sweep is contextual because it reuses the participant-averaged target and does not replace E008’s participant-level test.

C.5 E015 family, band, and partial-correlation sensitivities

E015 scored 22 decoder language models from six named families using tokenizer-independent bits per byte and the same controlled Tuckute alignment endpoint. Across the full capability range, the Pearson relationship was approximately $r = -0.78$. [E015] Removing near-floor models weakened the association: in the capable band it was approximately -0.50 , and in

the overlapping quality band approximately -0.48 . [E015] The Fisher intervals for these band-restricted correlations included zero because the bands contained only eight to ten models.

A partial correlation controlling for log parameter count was -0.675 with Fisher 95% interval $[-0.857, -0.344]$. [E015] This is not a clean decomposition into scale and quality because bits per byte and scale were strongly collinear and the residual variation also encoded family, data, and architecture differences. Family-cluster resampling remained negative, but the six families separated into roughly two model generations, limiting the effective independent breadth. Participant-level analyses showed the pattern in the one participant with adequate alignment signal and were uninformative in two participants near the measurement floor. The supported conclusion is that quality is a serious confound requiring matched-quality controls, not that perplexity deterministically explains brain alignment within the capable regime.

C.6 E017 intervention sequence and E023 identification gate

E017 tested whether full fine-tuning could reveal movement that low-rank adaptation missed. An aggressive configuration substantially worsened held-out language quality and failed the quality requirement. A gentler UTS03 pilot showed that the representation could move without the same failure, which cleared a three-participant existence test. Across UTS01, UTS02, and UTS03, the real-target arm did not improve over its target-specific controls. [E017] The observed subject-seed cells share only three biological units, so the result is reported descriptively as a three-participant mechanism probe rather than with a population confidence claim.

E023 then asked how an objective-specific loss could be separated from the alignment expected at the student’s quality. Its proposed estimand was the vertical residual of a KD student below a same-lineage alignment-quality frontier, paired against ordinary fine-tuning from the same initialization. The analysis-only slope gate found one locally flat lower-quality Pythia pair, but no Qwen pair met the predeclared saturated-regime requirement [E023]. Because the identifying region was not established, the training comparison was not launched. Together E017 and E023 distinguish a tested full-fine-tuning null from an untested larger-scale counterfactual program.

C.7 Compact notation

S	Language stimulus, such as a sentence or story segment.
X	Language-model feature block for the stimulus.
Y	Recorded or synthetic target response.
Z	Specified nuisance feature block.
ΔR^2	Cross-validated semipartial unique R^2 , the full-model R^2 minus nuisance-model R^2 .
$\mathcal{L}_{\text{brain}}$	Auxiliary loss relating a student representation to a neural or cognitive target.
$\mathbb{E}[Y \mid S]$	Stimulus-predictable conditional mean of a response.
ROI	Region of interest, a prespecified group or aggregate of voxels.
BPB	Bits per byte, a tokenizer-independent language-model quality measure on a fixed corpus.
KD	Knowledge distillation, training a student to match a teacher’s softened output distribution.

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